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Business Model Innovation for a Foodtech Startup

Solving Food Waste Using Real Time Spoilage Indicator

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Business Model Innovation for a Foodtech Startup - Solving Food Waste Using Real Time Spoilage Indicator

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Innovation av affärsmodeller för ett uppstartsbolag inriktat på livsmedelsteknik - En lösning för hantering av matavfall med hjälp av en indikator för matens nedbrytning i realtid

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Sammanfattning

Livsmedel, en väsentlig faktor för mänsklig existens, är en viktig orsak till negativ miljömässig och social påverkan, till följd av livsmedelshanteringens komplexitet, intressenternas arbetssätt och konsumenters beteende. Denna avhandling behandlar en livsmedelsprodukt, färskt kött, som har ett högt koldioxidavtryck på grund av omfattande mark- och vattenanvändning under dess produktion. Användningen av traditionella system för datummärkning, där det tryckta förfallodatumet skiljer sig från det datum då produkten är obrukbar, har lett till slöseri av denna resurskrävande produktkategori. För att hantera denna specifika fråga om avfall så utvecklar Innoscentia en färskhetsindikator som i realtid kan indikera matens kondition, genom att reagera på de organiska föreningar som frigörs när köttet bryts ned. Detta ska i sin tur minska klyftan mellan tryckt hållbarhetstid och faktisk hållbarhet. För att lyckas kommersialisera en sådan teknik måste företaget utgå från en välgrundad affärsmodell. Syftet med denna forskning är att förstå värdet av den ovannämnda tekniken och föreslå affärsmodeller som kan användas för att kommersialisera denna indikator. Detta görs genom att studera olika berörda intressenter genom litteratur, semistrukturerade intervjuer och en fallstudie av en baltisk detaljhandelskedja.

De huvudsakliga resultaten av den empiriska forskningen framför att den ovannämnda orsaken till att mängder av kött kasseras förekommer och att stora mängden avfall uppstår på grund av detta i detaljhandelsledet. De intervjuade intressenternas uppfattningar om matsvinn samt deras uppfattning om realtids indikatorn presenteras i avhandlingen. Med hjälp av dessa resultat så kan indikatorns värde presenteras för producenter, återförsäljare och slutkonsumenter. Slutligen presenteras fem olika affärsmodeller, baserat på olika värdeskapande möjligheter, med producenter som kundsegment. Dessutom presenteras en diskussion om återförsäljare som kundsegment och andra värdeskapande möjligheter. Studien avslutas med att ge rekommendationer till Innoscentia för att kommersialisera produkten och indikerar att ytterligare forskning bör göras på affärsmodeller för produkter som ger värde till flera intressenter i försörjningskedjan.

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Business Model Innovation for a foodtech startup - Solving food waste using real time spoilage indicator

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Abstract

Food, an essential to human existence is also a major contributor to negative environmental and social impact due to the complexity of the food system, current ways of working of the stakeholders and consumer behaviour. This thesis deals with one such food product, fresh meat which has a high carbon footprint due to extensive land and water usage during production. The use of traditional date labelling systems were the printed due date differs from the actual due date has led to wastage of such an impactful product category. To tackle this specific issue of waste Innoscentia is developing a real time freshness sensor which can indicate food spoilage by reacting to the volatile organic compounds released as meat shelf life decreases, thus bridging the gap between printed shelf life and actual shelf life. In order to commercialize such a technology in the complex food supply chain the firm has to have a sustainable business model. The purpose of this research is to understand the value of the said product and propose business models that can be used to commercialize this product. This is done by studying the different stakeholders involved through literature, semi structured interviews and a case study of a Baltic retail chain.

The main findings of the empirical research shows the existence of the aforementioned cause of fresh meat waste and presents the high magnitude of waste due this particular cause in the retail stage. The current practices, perceptions regarding food waste of the interviewed stakeholders are presented along with their perception of the real time spoilage sensor. Using these results the value of the sensor for the producers, retailers and end consumers is presented. Finally, five different business models with producers as the customer segment based on different value creation possibilities are presented. Additionally, retailers as a customer segment and other value creation possibilities are discussed. The study ends by providing recommendations to Innoscentia in order to commercialize the product and indicates further research to be done on business models for products specifically providing value to multiple supply chain stakeholders.

Acknowledgments

Every task accomplished however trivial it maybe would be only possible partly through human effort and partly through chance. The chance of things being the way they are, certain events in life and certain choices made based on intuition have led me to where I am today. I would like to show my gratitude to the cosmic energy that is perceived in different ways by many people for nudging me in the right direction and providing me with the opportunity and environment to learn and grow both professionally and personally.

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Abbreviations

BBD	Best before date
BMC	Business model canvas
BMI	Business model innovation
DC	Distribution Center
DSL	Dynamic shelf life
ERP	Enterprise Resource Planning
FSC	Food supply chain
FFSC	Fresh Food supply chain
FIFO	First In First Out
HACCP	Hazard analysis and critical control point
MAP	Modified atmosphere packaging
MTO	Make to order
MTS	Make to stock
TTI	Time Temperature indicator
VOCs	Volatile organic compounds

1. Introduction

1.1 Background

Food, an essential for human existence, is also a cause of environmental and social problems, not by itself but due to the ways the complex food system works and consumers behave (Lundqvist, de Fraiture and Molden, 2008). It is estimated that around 1.3 billion tons of food is wasted every year on a global scale which is around one third of the produced food. Food is wasted throughout the whole supply chain from producers, distributors, retailers to consumers in different ways and quantities. In the EU alone, 88 million tons annually is estimated to be wasted. One of the many reasons food is wasted is due to the current date labelling i.e. food considered inedible as it has passed the expiry date is a major cause accounting to over 50% of the waste (Göransson, 2019; Kaipia, Dukovska-Popovska and Loikkanen, 2013). Food waste is not just about the economic loss (Hanson and Mitchell, 2017) but also an environmental impact as depleting resources such as land, water and energy are used to make this food. Estimates show that 18 million tons of CO_2 are generated due to throwing away edible food in the UK (Lundqvist, de Fraiture and Molden, 2008). This shows a need for the current stakeholders in the food supply chain (FSC) to focus on sustainability for the betterment of the environment and as a business opportunity.

Highly perishable food items such as fresh meat, fish, dairy, fruit and vegetables are the majority of the above mentioned waste as these have low shelf life. These foods are also being increasingly consumed as the income levels in BRIC economies increase, which means potential for higher amounts of waste unless changes to present situations are made (Parfitt, Barthel and MacNaughton, 2010). The stress on food safety can be understood from strict regulations and huge recalls (Lundqvist, de Fraiture and Molden, 2008), the needs of today's Consumers are also evolving, the increase in demand for safe, healthy and low environmental impact food can be seen (McCarthy, Liu and Chen, 2016). This puts more stress on the FSC stakeholders to consider sustainability as a competitive advantage and brings the need for sustainable solutions in the FSC.

Innoscentia is a startup located in Sweden developing one such solution to enable FSC players to achieve their sustainability goals. Targeting the fresh meat market and the problem of date labelling above mentioned, Innoscentia is developing a low cost chemical sensor to detect meat spoilage real time, aiming to replace the existing date labelling systems. This company and the technology will be the main focus of this research. It is evident from Chesbrough and Rosenbloom, (2002) and Teece, (2010) that technologies alone are not enough to create a successful business but a business model is required to bring out the latent value of the

technology. Business models are required in order for a technology to be commercialised and to build a sustainable competitive advantage and profit for its innovators. Business model is the logic through which a firm creates, delivers and values to customers and the way it captures value for itself in the process of value delivery. It also includes defining what the value is for different customers. Not only are they a requirement but another way for a firm to innovate i.e. business model innovation (Teece, 2010). While this provides enough motivation for new firms like Innoscentia to explore business model innovation, the industry they are located in provides further motivation. The FSC is complex and filled with many different actors, with different levels of power concentration among players (Kähkönen, 2011). Technologies like that of Innoscentia's can provide different values to different stakeholders in FSC, this makes development of the appropriate business model a necessity and a major challenge (Göransson, 2019).

1.2 Purpose

The purpose of this research is to propose sustainable business models for Innoscentia to commercialise their analogue sensor for detecting fresh meat spoilage. This includes understanding of different components of the business model mainly, the value of such a technology to different stakeholders in the meat industry, the present state of the meat supply chain and characteristics of these stakeholders to design the value delivery and value capture mechanisms.

1.3 Delimitations

As an entrepreneurial firm which was in a complete R&D phase previously and now moving to the stage of releasing their first fully working prototype, Innoscentia has many questions unanswered in terms of the product and the business. This means assumptions are made and presented in the research when required.

The business model concept as will be presented in the frame of reference chapter is vast and complex. Therefore there is a limit to the amount of detail and the level of aggregation to which a business model can be developed in this study. Furthermore, in relation to the current state of the company certain components of the business model are discussed more than the others. Furthermore, the focus of the study is on the Swedish market.

1.4 Disposition

Chapter 2 - Frame of reference - This chapter presents an overview of the business modelling from a theoretical perspective. Definition of business model and business model innovation are discussed. Furthermore, the framework used in this study for sustainable business model design is presented.

Chapter 3 - Industry Context - This chapter presents multiple topics whose understanding can be considered prerequisites to validate and generate new business models. The fresh meat supply chain, date labelling and legislation, food waste, food spoilage sensor market are presented using existing literature.

Chapter 4 - Methodology - This chapter presents the research setting by introducing Innoscentia and the present state of the company and technology. Followed by a detailed description of data collection and analysis including motivation towards chosen methods.

Chapter 5 - Results and Analysis - This chapter is presented in two main parts. Firstly, empirical results from the collected data are presented in relation to the industry context by showing similarities and differences to extant literature. Secondly, business models developed for Innoscentia using the knowledge gathered are presented.

Chapter 6 - Discussion and Conclusions - The final chapter builds on the overall research and presents the main findings. It provides recommendations to Innoscentia and finally limitations to the present study are presented.

2. Frame of reference

2.1 The Concept of Business Models

Understanding of business models has come a long way since the first use of the term in 1957, with an extensive increase in use both by practitioners and researchers starting from the mid 1990s this increasing trend continues in recent years (Massaécole, 2017; DaSilva and Trkman, 2014). While this increase can be attributed to emergence of internet technologies which provided new ways of doing business, focus on bottom of pyramid issues, increase in exploring ecological and social sustainability as core values and value propositions, it also means that researchers have studied the concept using different lenses leading to many different definitions and interpretations of the term (Zott, Amit and Massa, 2011; Massaécole, 2017; Massa and Tucci, 2013). However Zott, Amit and Massa (2011) identify that there are emerging themes across these different definitions that suggest a business model as a unit of analysis, a system level approach to explaining how firms do business, the important role of activities spanning the organisational boundaries and the focus on both value creation and value capture. These different interpretations include varying amounts of content i.e. certain definitions have considered only revenue models as the unit of analysis whereas some have considered the activity systems to create and capture value. Definitions explaining the same content have also used different terminology creating further confusion and moving away from a consensus on the definition of business model (Zott, Amit and Massa, 2011; Massaécole, 2017). Therefore, instead of debating the exact definition of a business model this research presents what it considers the definition of a business model to be and bases the rest of the work on the same.

Business model is a formal conceptual representation of the way a firm does business. It describes the value proposition, value creation and delivery and value capture mechanisms of the firm.

Business model is a formal conceptual representation of the way a firm does business. It describes the value proposition, value creation and delivery and value capture mechanisms of the firm. In other words it describes What the value is? Who is the value for? (value proposition) How is this value generated? Who generates this value? How does this value reach the customer? (value creation and delivery) How does the firm generate revenue? What are the costs associated with value creation and delivery? (value capture) (Teece, 2010; Osterwalder and Pigneur, 2010; Massaécole, 2017). Firstly, this definition takes the systems perspective and considers multiple components and their relations (Zott, Amit and Massa, 2011). Secondly, this choice of definition

clarifies that the business model is not considered as a cognitive/linguistic schema or as an attribute of a real firm but as a formal representation. Cognitive/linguistic schema interpretations such as "stories that explain how an organisation works" (Magretta, 2002) simplify the notion of business model and leads to ignoring relevant information or the lack of it creating inertia to the use of models or leading to incorrect decision making. Attribute of real firm interpretations such as business model is a set of interdependent activities the firm chooses and performs with the use of selected resources and capabilities to create value (Zott and Amit, 2010) describe how firms do business whereas the chosen interpretation gives a formal conceptualization of the same. The formal conceptual representation interpretation can fulfil the functions of the other interpretations such as communication of business models across the company and stakeholders, discussing impact a model can have on organisational performance while fulfilling the function of studying focal elements and their dynamics (Trimi and Berbegal-Mirabent, 2012; Massaécole, 2017). Furthermore this definition includes all the recurring themes presented by Zott, Amit and Massa, (2011) implying that most research will arguably agree on this definition. Additionally this definition can be used across multiple subject matter lenses as classified by Zott, Amit and Massa (2011) into e-businesses, strategy and technology and innovation management.

Business models in reality are complex phenomena, the use of formal conceptual representation allows us to deal with such complexity by simplifying it. Which means that these representations are abstract and do not necessarily contain every detail, rather they focus on what they consider the most important components. This creates a possibility for researchers to decide on the level of abstraction and the content of the model i.e. what they consider to be important and excludable. Additionally the way it is represented can be different i.e. the semantics (Massaécole, 2017). Massa and Tucci (2013) present the different levels of abstractions as narratives, archetypes, graphical frameworks, meta-models and activity systems where narratives are farthest from reality i.e. more abstract and activity systems the closest i.e. less abstract. While narratives and archetypes allow for sense making of the business model they are too simplistic to use for experimenting and studying models in depth. On the other hand meta-models and activity systems are highly detailed and help represent the dynamics between different components. However, they are more arduous especially considering the generation of multiple models to experiment and the firm in focus, Innoscentia, is still in the initial stages meaning there are many things not known at a detailed level as presented in the delimitations. Additionally meta-models such as Casadesus-Masanell and Ricart (2007) and activity systems such as Zott and Amit (2010) which focus on business models as a set of choices & consequences and as a set of interdependent activities respectively do not explicitly represent the notions of what value is. Arguably this is perfectly alright for incumbent firms looking to reconfigure the business model, which needs clear understanding of the existing dynamics, whereas the understanding of value is equally important as the dynamics between components for new firms (Blank, 2007; Massa and Tucci, 2013). Though the dynamics are not explicitly presented in the graphical representations,

they are considered while building and experimenting with the models meaning there is more depth to graphical representations than what meets the eye (Osterwalder and Pigneur, 2010). Description of these dynamics between components in a model is given when required in the results and discussion chapters of this thesis.

Following in the lines of the definition of a business model, there is heterogeneity in literature regarding the components of a business model representation (Massaécole, 2017). In their study Morris, Schindehutte and Allen (2005) mention the use of 24 different components in literature, many of these are mentioned by multiple researchers who study e-commerce businesses and businesses in general. Nenonen and Storbacka (2010) also present similar results, it is also evident that different terminology is used to explain similar components. The business model canvas (BMC) is a representation presented by Osterwalder and Pigneur (2010) which contains nine interrelated components, viz. Value proposition, Customer segments, Channels, Customer relationships, Revenue streams, Key resources, Key activities, Key partnerships, Cost structure. These nine components are not only very much in line with the selected definition, but are also the most mentioned components in literature (Morris, Schindehutte and Allen, 2005; Nenonen, S. and Storbacka, 2010). It is also necessary to understand that components other than these nine presented by researchers are important but these are considered not a part of the business model or as outcomes of a good model design and implementation (Osterwalder, 2004). Take for instance the competitive strategy which is considered a component by Chesbrough (2007), this is rather a part of designing a strong business model but not part of the model itself. Similarly when studies such as Morris, Schindehutte and Allen (2005) and Nenonen and Storbacka (2010) use more specific levels such as design principle and rules in their frameworks these are not new components of the model. Based on the above reasoning this holistic representation which suits the needs of this research i.e. the BMC as shown in figure 1 is chosen to study and generate the business models for the focal firm. The nine interrelated components are now presented based on Osterwalder et al. (2014), Osterwalder (2004) and Osterwalder and Pigneur (2010).

1. Value proposition

It is the set of products and/or services which the focal firm offers to a specific customer segment, these offerings fulfil customer's needs and are a way for the firm to differentiate from its competitors. As the needs of a customer segment can differ from one another a firm can offer different values to different customer segments at the same time. Value for the customer can be created in different ways, it could be by fulfilling a functional need such as transportation or heating a room. It could also be something which reduces customers' risks and addresses their pains such as insurance contracts or waiting in queues for long times. At the same time the value could be something which creates gains for the customer while using the product or service such as creating economic savings.

2. Customer segments

A group of customers which have similar needs or act in a similar way can be made into one customer segment. A firm not only segments its customers but also chooses to serve particular segments and ignore others based on goals and criterias such as profitability and business sustainability. A segment can be created in multiple ways depending on the industry, for example segments in banking can be customers with huge sums of money and requiring ease of transactions and customers with smaller sums of money and need for savings. Similarly in the automobile market customers segments can be customers looking for economy cars strictly for transportation and customers looking for sport cars as an identity. In B2B businesses there are several stakeholders involved in the process of transacting the value proposition, their involvement changes the purchase decisions. In such B2B transactions the end user can be either active or passive in the decision making whereas other stakeholders can be influencers, recommenders, economic buyers controlling the budget and saboteurs who can obstruct the process or the decision for various reasons main one being losses such as economic and brand identity for them.

3. Channels

The connection between a customer segment and the firm's value proposition is created through channels. Marketing, sales and distribution channels are the different channels useful at different stages of the customer buying cycle from creating awareness of the product and communicating the value, evaluating and purchasing the product to after sales service. These channels are important for customer experience and firms aim to provide best possible convenience mainly depending on the value they want to provide and the costs associated with it. Multiple channels can be chosen by a firm based on their goals, these channels can be direct or indirect both having their advantages and drawbacks. Some of the most used channels are retail stores, websites, and sales personnel. Not only choosing the best channels is important but looking for channel conflicts is equally needed. Incumbent firms often looking to innovate their business model choose to opt for new channels such as website sales which conflict with their existing channels such as dealerships which have been built on long term relationships. The firm's choice to handle or avoid these conflicts generally depends on the long term strategy.

4. Customer relationships

Every firm creates relationships with its customers whenever there is an interaction, the kind of relationship it creates depends on the customer segment it is targeting and the cost to build this relationship. These relationships can be classified into acquisition, retention and upselling. Firms use different mechanisms to maintain these three kinds of relationships, these relationships are interrelated. For example a choice made to increase customer acquisition through discounts can have an effect on customer retention as these acquired customers can be very price sensitive.

These relationships are built through the channels at different stages of the customer buying cycle mentioned previously, this shows the interdependence of the two elements.

5. Revenue streams

The translation of the value offered into monetary value for the company is done through the revenue streams. A firm chooses to create revenues from the values each customer segment is willing to pay for. Multiple revenue streams each with a different pricing mechanism are also used by firms. On a general level these streams can be classified as one-time and recurring revenues. Some of the most followed revenue generation mechanisms are sale of assets, subscription fees on services, leasing assets, licensing and advertising. These mechanisms can have different pricing strategies such as fixed and dynamic pricing.

6. Key resources

In order to generate value for their customers firms need different types of resources. These range from tangible resources such as manufacturing units, equipment, building, etc. Intangible resources such as brands, copyrights and patents, data. Human resources i.e skilled workers who create knowledge and use the other resources to create value. Financial resources to support different pricing systems and to buy the new resources. The selection of resources is based on the value that needs to be created, the channels and relations needed to be maintained with the customer. These sets of resources provide the capabilities to a firm, which can help them be competitive. However resources are not the only ones which provide capabilities, for instance multi sided platform business models in themselves provide new capabilities for the firm due to aggregation of both buyer and supplier side of the industry. Furthermore capabilities can be outsourced.

7. Key activities

A wide variety of activities are performed in order to create, deliver and capture value. Primary activities are the ones directly creating the value proposition and its delivery. Some of the main primary activities are production, problem solving and network creation. Secondary activities such as human resource management, infrastructure building support these primary activities and are used generally by every firm. Activities are linked to resources i.e to certain resources are needed to perform certain activities. These activities can also span the organisational boundaries, this brings us to the next element, key partnerships.

8. Key partnerships

Partnerships of different kinds with different purposes can be formed between firms. Supplier partnerships build reliable flow of material needed to create the value. While key resources and key activities can be outsourced from strategic partnerships with both competitors and non

competitors. These are done to reduce risk, uncertainty, reduction of costs, learning and knowledge creation among various other reasons.

9. Cost structure

A cost exists to perform every activity and acquire the required resources and partnerships. It is only after the subtraction of these costs from the revenue a firm sees its profit or loss. The cost structure is a way to understand what the most expensive activities are in turn showing where a business can reduce its costs. Both cost driven strategies where making least costs possible is the goal and value driven strategies where more focus is on value creation are used by firms. A way to classify costs is fixed and variable costs.

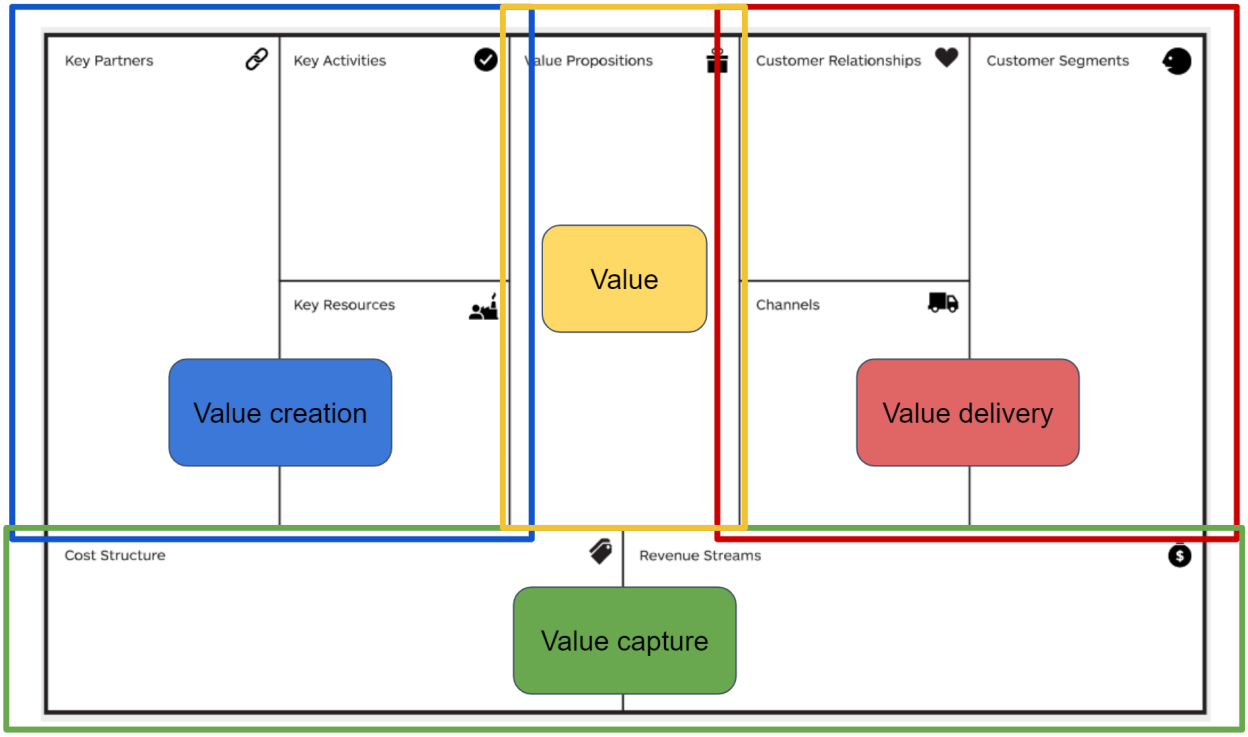


Figure 1. Business model canvas and its relation to the chosen definition

2.2 Business Model Innovation

Novel changes and/or design of the elements of a business model and/or the architecture linking the elements of the business model can be defined as business model innovation (BMI). Business models can be viewed as a vehicle of innovation i.e allowing commercialization of new technologies. They can also be viewed as a source of innovation and competitive advantage. It is of course possible that both of these phenomena occur at the same time such as in the case of Xerox where the deployment of the first photocopy machine was done through a novel model

involving leasing (Foss and Saebi, 2017; Massa and Tucci, 2013). Additionally, BMI can be seen as a different phenomenon in new or entrepreneurial firms and in incumbent firms. While new firms are in the search of and designing a fitting business model, incumbent firms reconfigure existing business models. These two activities face different challenges. New firms or incumbent firms venturing into new businesses face challenges such as higher amounts of uncertainty in terms of customer response and market conditions, whereas incumbent firms face constraints due to path dependency of existing models and cognitive barriers due to dominant logic (Massa and Tucci, 2013). However the complexity of BMI exists in both types of firms. This complexity is two fold, (1) the possible combinations of different elements of the business model is large, (2) the non linear interdependencies between the components. Though this complexity exists, extant literature shows that innovation can occur in both or only one dimension of this complexity based on the nature of the unit of analysis. Thus, allowing innovation either in individual components of the model or combination of components and/or in the entirety of the model. Furthermore this BMI can be something which is new for the firm or new for the industry/world (Foss and Saebi, 2017). Another dimension is that of the intensity of innovation i.e BMI can be either incremental or radical innovation (Wirtz and Daiser, 2017). Figure 2 shows these three dimensions of BMI.

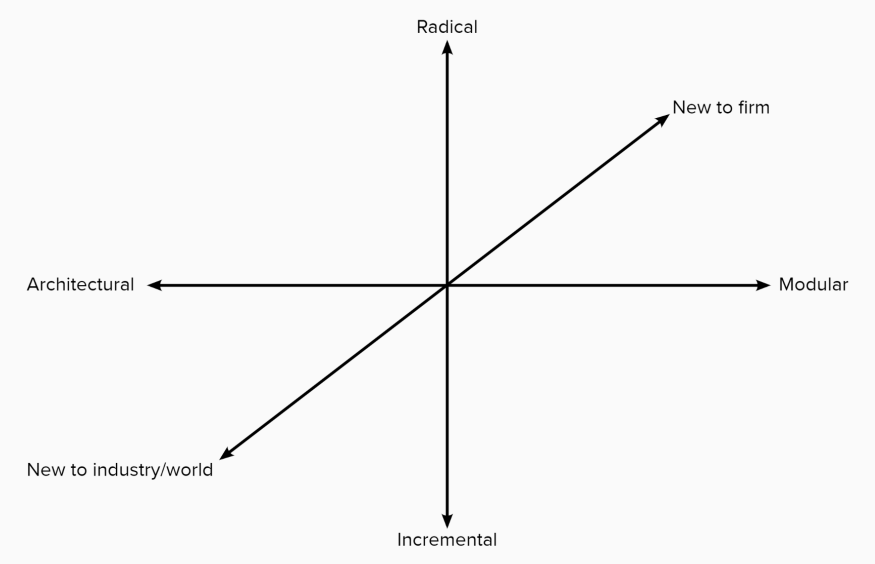


Figure 2. Business model innovation dimensions. Scope of innovation and Intensity of innovation presented on horizontal and vertical axis respectively. Degree of innovation presented on the third axis.

This level of complexity and the uncertainty related to markets and customers more so in entrepreneurial firms evidently suggest that fully planning business models *ex ante* is not possible. Moreover, finding a viable and perfect business model is a result of experimentation, learning and adjusting during which many failures of proposed models are also encountered (Teece, 2010; Massa and Tucci, 2013). This experimentation not only allows us to evaluate the

existing assumptions but also leads to effectuation leading to discovery of new information that would not have been available otherwise (Chesbrough, 2010). While such experimentation with the real world is out of the scope of this research, experimenting by building multiple business models (Chesbrough, 2010) i.e with different combinations and architecture of the elements and analysing them both qualitatively and to a certain extent qualitatively is done, which is presented in the Results chapter. Furthermore as Teece (2010) presents, building these models needs understanding of multiple stakeholders, customers and competition. This information regarding multiple stakeholders is presented in the Industry context chapter and the empirical research performed provides additional information regarding customers. Another factor to consider is that BMI is not only discovery driven but it also needs to be done continuously to attain dynamic capability, both incumbent firms and new firms which have found a viable model reach a stage where the competitive advantage wears off. This could be due to new technological, organisational innovations or even changes in customer needs. This means a business model can only be sustainable for a certain amount of time, that amount however is contextual (Teece, 2010; Foss and Saebi, 2017). This leads to exploration of the question "What is an effective business model ?".

While BMI can be done in order to achieve specific goals by following different design themes such as novelty, cost differentiation (Zott and Amit, 2010) or by following archetypes such as freemium, razor-blade, subscription models (Osterwalder and Pigneur, 2010) there are two general goals that every model needs to achieve in order to be a effective and provide sustainable competitive advantage which are (1) Configurational fit (Nenonen and Storbacka, 2010)(Morris, Schindehutte and Allen, 2005; Frankenberger et al., 2013) and (2) balance between non-imitability and flexibility (Foss and Saebi, 2017; Teece, 2010). There are two parts to the configurational fit : internal configurational fit and external configurational fit. As the components of a business model can affect and be affected by other components, an internal configuration fit means that these components have a consistency and are being orchestrated in a manner to have no conflicts. Furthermore, if these components can reinforce each other a higher degree of internal configurational fit can be achieved. The external configurational fit is to do with the alignment of the business model with the business environment i.e with industry forces such as suppliers, stakeholders and competition, trends such as regulatory, cultural and technological, market forces such as switching cost and changing needs and macro economic forces such as global market conditions (Nenonen and Storbacka, 2010; Morris, Schindehutte and Allen, 2005; Osterwalder and Pigneur, 2010). While external fit can be pursued by performing the first step of understanding the environment, mainly the needs of customers and behaviours of stakeholders, internal fit can be pursued in the next step while designing the models by actively ensuring alignment and possible reinforcements (Frankenberger et al., 2013). These reinforcing elements also create a barrier to imitation. However, even with high configurational fit business models are subjected to replication or imitation, this can be seen as

many successful business models which have performed well or disrupted industries have been replicated by competition and have also been used as archetypes in the long run. However, it is important to understand that success of a model does not lie in design alone. Perfect implementation which requires system and processes, social complexity and path dependency also create barriers to imitation (Teece, 2010). Though designing reinforcing components to create a competitive advantage might sound like the perfect way, the factor of business environments being subjected to inevitable change needs to be taken into account. This means the competitive advantage created will wear off as the external configurational fit is lost as a result of change (Foss and Saebi, 2017). Furthermore, firms need to adapt their business models to the changing environment by changing certain components which will affect the internal configurational fit previously achieved and attaining the fit needs to be pursued again. However, this time attaining fit is also affected by the previous choices i.e path dependency, models with tightly knit elements are difficult to change compared to models with rather loosely knit elements (Morris, Schindehutte and Allen, 2005). This is due to the interdependencies between elements which are also dynamic, creating a high level of complexity (Demil and Lecocq, 2010). Therefore, firms need to find the balance between non-imitability and flexibility while designing their business model in order to be able to adapt to change (Foss and Saebi, 2017; Teece, 2010).

2.3 BMI in Supply Chains

Often consisting of many different members with different levels of importance, supply chains are complex systems through which the value, product, and information flows in order to keep the end consumer satisfied. The horizontal structure of a supply chain would represent the number of tiers in the chain and the vertical structure represents the number of suppliers or customers in each tier. Each member is connected with different activities which are of different levels of importance as some would require more resources and management than others. The importance of these activities would differ from the perspective of each member of the chain. Changes to these business processes or activities aimed towards generating a specific customer value, the structure of the supply chain network and technologies that can be applied individually or in combination with other changes create supply chain innovation. These three changes can occur in different degrees such as radical or incremental innovation and intra firm or inter firm level (Arlbjørn, de Haas and Munksgaard, 2011). Supply chain innovation and BMI go hand in hand, further supply chain innovation can be considered a subset of BMI or it can be considered as a way to generate BMI. The parts of BMI i.e the changes to value creation and value delivery are very similar to that of or are achieved by changes to the three elements i.e process, structure, technology of a supply chain. On a more detailed level changes to the supply chain can be done to achieve supply chain innovation by adding, eliminating, relocating, dividing, accelerating, inverting or parallelizing activities which would call for changes in the structure and members of the chain. Both supply chain innovations and BMI are dynamic in nature and have the ability to

trigger one another and on an overall level they are triggered by interaction with the business environment (Abdelkafi and Pero, 2018; Arlbjørn, de Haas and Munksgaard, 2011). The activities related to supply chain management and innovation on an abstract level are to provide superior end customer value however, when seen on a more concrete level they aim to improve the effectiveness and efficiency of the supply chain. Improved efficiency, reduced costs, better customer service and reaching customer demands, serviceability, relationship management are some common measures of supply chain management and innovation. These are achieved by various innovations and technologies such as collaborative forecasting, continuous replenishment program, just in time production and sourcing, tracking and tracing technologies, digitalisation using IT and IoT systems (Arlbjørn, de Haas and Munksgaard, 2011; Abdelkafi and Pero, 2018).

The impact of the aforementioned technologies and other supply chain innovations can vary from a single business unit in a firm to multiple stakeholders in the supply chain. This difference in impact affects the way a technology is valued by different stakeholders in turn indicating the way it can be deployed and diffused in the supply chain (Kim, Cavusgil and Calantone, 2006). Technologies such as Enterprise Resource Planning (ERP) systems are designed to integrate different supply chain processes of a firm's functional units. These allow for planning and forecasting more efficiently with fewer mistakes. Such systems are intra firm level and provide value to the stakeholder who employs it. However, interfaces can be created to integrate both suppliers and customers to this system making it more valuable. The company offering these ERP systems can use the network effects created, the more the number of stakeholders using such a system the more the value of it increases (Dawson, 2002). In the case of technologies such as RFID tags the traceability of the product is increased and the process for the players down the supply chain are made easier. A major difference however is that the stakeholders deploying the solutions do not necessarily gain value from these technologies as much as their customers do, i.e a major difference in the value chain compared to that of ERP systems (Sarma, 2012). This basically leads to three categories for the supply chain technologies. Category 1 is where only the stakeholder of the supply chain deploying the technology gains value. Category 2 is where both the stakeholder deploying the technology and one or more other stakeholders of the supply chain gain value. Category 3 is where the supply chain stakeholder deploying the technology does not gain value but provides value to one or more other stakeholders in the supply chain. Growing digital technologies can come under any of these based on the amount of technology that is deployed, for example using an auto checkout system in a store mainly improves processes of that store alone, however, the barcode system itself is used both by the distribution centres and all other retail outlets equipped with scanners. In addition to these categories the producer of these technologies may or may not be a part of the same supply chain. While most digital technologies are not part of the same supply chain, packaging technologies and its producers are integral part of the supply chain mainly in the food industry. Food producers are deployers of such technology as will be shown in the next chapter and the value can be gained by multiple

stakeholders depending on the packaging technology.

Packaging innovations can be studied as a separate topic altogether including its effects on marketing and product differentiation, however studying it in a lens of supply chain innovation provides insight into its diffusion. Food packaging mainly consists of the primary packaging which is in contact with the food and is the packaging mainly presented to the end customer. Secondary and tertiary packaging of food is mainly aimed at protecting food and the primary packaging during distribution, easier handling of products aimed towards operational efficiency. Some secondary packaging is also used as a key marketing space for the end consumer. The design of primary packaging is majorly determined by the end consumer needs; these are quality of food, shelf life, safety, convenience and variety. While these are the functional needs of the end consumer, primary packaging also caters to product differentiation by providing product differentiation in terms of brand image and cost (Coles, McDowell and Kirwan, 2003). The drivers of packaging innovation can be both the needs of the consumers and the technological development which of course to be commercialised successfully need end consumer acceptance (Syvänen, 2020). Therefore, the primary packaging value chain is mainly based on the end consumer acceptance. Similarly needs such as durability, reusability and cost effectiveness are the drivers of secondary and tertiary packaging. These however provide value to the retailers and logistic companies, hence the value chain and implementation depends on their acceptance. Packaging innovations such as the intelligent packaging and active packaging focal to this research tend to provide different amounts of value for all stakeholders of the food supply chain. Such innovations require changes at each stakeholder of the supply chain inorder to be successful, additionally, the value for each stakeholder might not be quantifiable completely(Gustafsson et al., 2006; Coles, McDowell and Kirwan, 2003). Change in a supply chain is a complex process due to the number of stakeholders involved and the power dynamics between these stakeholders. These stakeholder companies can be classified into integrative or segmentalist. Integrative companies are the ones with more collaboration, open communication, holistic problem solving and top management support for innovation and change. Whereas segmentalist companies are more conservative and traditional where people work in silos with lower collaboration, support for new ideas is lower and these companies generally have the lower ability to manage rapid and emerging changes in the market. A supply chain would consist of stakeholders of both these types with different amounts of power. Change is easier to implement and accomplished faster in supply chains where dominant players are integrative. The opposite is true in supply chains with dominant players being segmentalist (Gustafsson et al., 2006). This supply chain view of packaging innovation is of utmost importance for the companies developing these innovations as the implementation, strategies to go to the market and the business model would be affected by the position of the dominant players in the supply chain and their acceptance of the innovation.

The extant BMI literature identifies the dimensions of interaction with the business environment both at a macro and micro level, part of is the industry and market shifts (Wirtz and Daiser, 2017). However, not many frameworks have taken this into account on a more explicit scale, the BMC framework used in this research as well considers these interactions behind the scenes so to say. It can also be seen in literature that most BMI innovation studies are done on a theoretical and general scale with few studies focusing specifically on specific industries, markets and firm contexts. These specific studies as well have not focused on BMI in supply chain contexts as described in this section. Reviews show the need for future research on BMI elements for specific contexts and the tools that can be used for practical applications (Schneider and Spieth, 2013; Foss and Saebi, 2017). Interaction and collaboration with stakeholders has been identified as necessary in order to perform sustainable BMI mainly in the areas including ecological and social sustainability. The sustainable BMI literature could therefore provide insights into more ways of including stakeholder interaction in general BMI. However, it is to be noted that this increasing area of research is new and might not already provide applicable BMI tools (Brennan and Tennant, 2018; Geissdoerfer, Vladimirova and Evans, 2018). Finally, in the current BMI research Casadesus-Masanell and Ricart (2012) present a way to evaluate business models in isolation and with respect to the other player's business models in the environment that the business model would interact with. They explain the possible interdependency of business models and classify possible interactions into tactical and strategic interactions which can be used to intensify or reduce the interdependencies. The framework of the business model given by Casadesus-Masanell and Ricart (2012) is on a more detailed level compared to that of BMC but does not have the same visual convenience as a BMC. While combining it with the current BMC framework might need more theoretical work it can still be used in the thought process of building the canvas.

3. Industry Context

3.1 Meat Spoilage and Shelf life

Fresh raw meat and poultry are considered highly perishable food. The spoilage mechanism of the meat depends on its source (beef, pork, chicken, etc.) and the composition of meat. Microbial spoilage, lipid oxidation and autolytic enzymatic spoilage are the various reasons for spoilage (Dave and Ghaly, 2011). The main factor determining the quality and safety of meat is microbial spoilage, these microbes are present on the meat surface and more could be introduced by contamination during the production process of meat (Lambert, Smith and Dodds, 1991). Microbial spoilage occurs due to the growth of various types of microorganisms like bacteria, yeasts and moulds. While yeasts and moulds are not predominant spoilage microbes, different types of bacteria are the main contributors to spoilage (Sperber and Doyle, 2009). While pathogenic bacteria such as Listeria, Salmonella, Clostridium and E-coli which are health risks are controlled by hygiene and decontamination in the production process. The growth of spoilage bacteria can be slowed by different mechanisms such as modified atmosphere packaging (MAP) and vacuum packaging. Methods such as irradiation, antimicrobial dipping, hygienic production conditions are used to reduce the initial microbial load inturn slowing the microbial growth (Lambert, Smith and Dodds, 1991; Petrou et al., 2012).

The Shelf life of the meat product depends on the growth rate of the microflora, which in turn mainly depends on the storage conditions such as temperature and the type of packing. The dominant microbes causing spoilage also depend on the type of packing which mainly changes the availability of oxygen causing aerobic or anaerobic conditions (in vacuum packaging and MAP) (Sperber and Doyle, 2009). In refrigerated meat products under aerobic conditions Pseudomonas, Acinetobacter and Enterobacteriaceae are the main microflora associated with spoilage, whereas under anaerobic conditions Lactic acid bacteria and Enterobacteriaceae are more predominant. The human perceivable indications of spoilage are mainly off- odour development and slime formation, which are caused by these microflora as an end product of metabolism (Lambert, Smith and Dodds, 1991; Sperber and Doyle, 2009).

The determination of shelf life of meat products is done to figure out mainly the quality over time and the safety over time. Sensory evaluations are done to assess the meat quality (taste and texture) using sampling procedures over a period of time (Jonsson, 2012). The spoilage time can be determined by various microbiological study methods which follow the principle of determining the amount of spoilage bacteria present at different intervals of time, some of which

have been practised since a long time and some of which are new methods such as Proton Transfer Reaction Mass Spectrometry to measure Volatile Organic Compounds (VOCs) (Jonsson, 2012; Mayr et al., 2003). Since there is possibility for temperature changes across the supply chain, shelf life calculations include safety margins which are currently set arbitrarily, this brings the need for newer methods to predict life by taking dynamic situations into consideration (Corradini, 2018). This difference in the estimated life and actual life have been shown through experimentation (Göransson, Nilsson and Jevinger, 2018). Both situations where estimated life is lower than actual life due to higher margins and no cold chain breakage causing food waste and estimated life being higher than actual life due to breakage in cold chain causing food safety risk exist. Addressing the same issue real time shelf life applications using temperature monitoring are being developed to make better predictions and provide that information in real time which need sensors across the supply chain and data gathering systems (Göransson, 2019; Yimenu et al., 2019; Jonsson, 2012). The analogue Innocentia sensor falls into this category of real time spoilage detection but provides a more direct measure of microflora (by measuring VOCs) compared to that of temperature based estimations.

3.2 Fresh Meat Supply Chain: Characteristics, Actors and Activities

Supply chains in general are complex consisting of various actors performing diverse sets of activities to produce value. Each actor of course tries to optimise their activities and accomplish economic, environmental or social goals. Fresh food supply chains (FFSCs) are even more complex and difficult to manage due to the product and market characteristics, these even make responsiveness and flexibility the supply chain requirements. Fresh food products are perishable. i.e quality deteriorates over time meaning a short shelf life for the product. This product characteristic hinders the applicability of inventories as a buffer and gives rise to market characteristics such as demand for short lead times. Another product characteristic is that of demand variety i.e products variants with similar characteristics like products in different package sizes and types. Additionally each variety has a different demand inducing volume variability. On top of all these complexities there is the market characteristic of demand uncertainty. This uncertainty deals with the prediction of demand which is challenging due to instability of demand which is generally attributed to season and weather. This prediction becomes even more challenging when promotions of products are included, as demand becomes more uncertain (Romsdal et al., 2011; Kaipia, Dukovska-Popovska and Loikkanen, 2013). Furthermore, these products are also the main cash flow contributors or profit generators for retailers making them a focal product for which retailers strive to maintain quality and permanent on shelf availability (Teller et al., 2018). Keeping these complexities in mind the structure of the fresh meat supply chain and the actors at different stages of it are now explored.

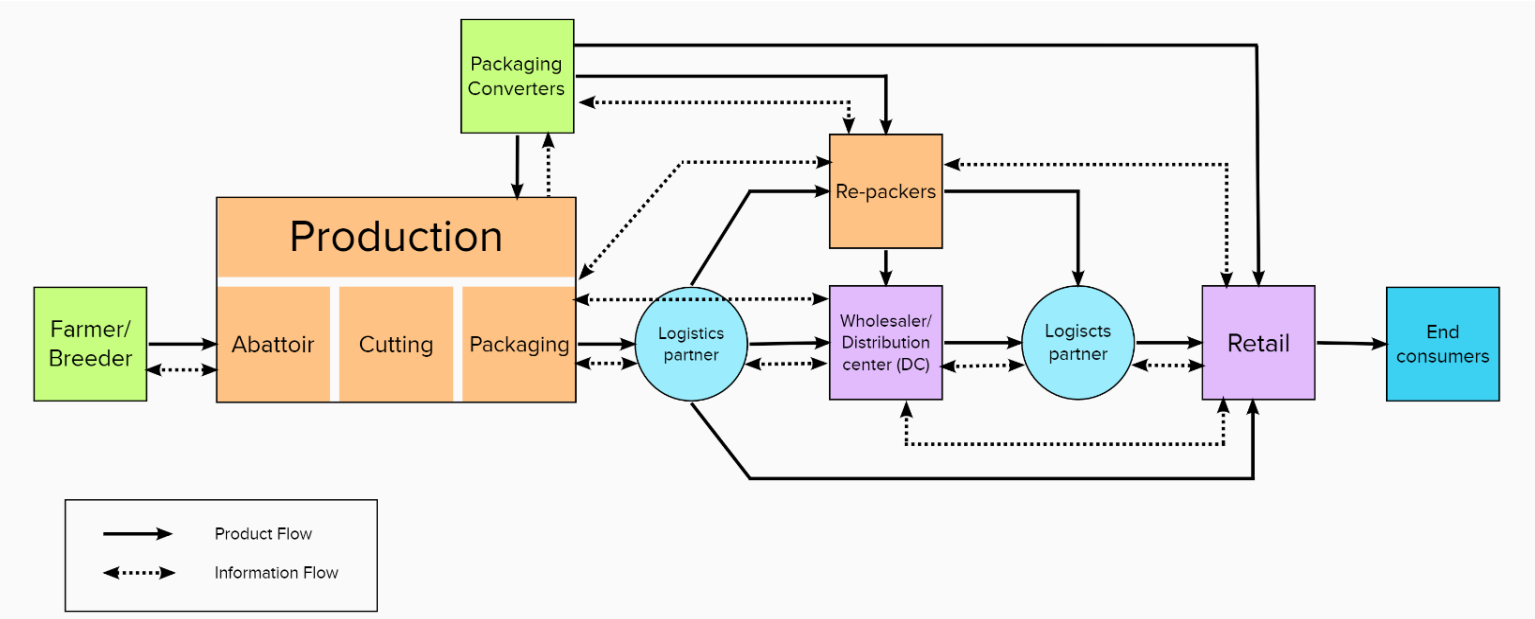


Figure 3. Structure of fresh meat supply chain

The generic structure of the fresh meat supply chain along with the product flow and the information flow is presented in Figure 3. Starting from the farmers where animals are bred to the end consumer where the meat is eaten the product undergoes different processes and changes hands of different players from farm to fork. Once the framers deliver the animals to the abattoir, they are slaughtered and drained of blood to be ready for cutting. Depending on who the product is for i.e industrial use or end consumer use, the slaughtered animal is cut into different sections at the cutting plant, which is then packaged based on the use. As transporting a slaughtered animal is cheaper and easier than live cattle, abattoirs are generally located closer to the farms. Imports and exports of meat also start to happen only after slaughter. Once packaged these goods are sent to the wholesaler through logistics partners, the whole supply chain is a cold chain where the temperatures are kept low to maximise the product shelf life. Wholesalers procure products from different producers including repackers who process and pack meat mainly for private label products. All these products reach the distribution centres from where they are sent to retail stores through logistic partners to reach the end consumer (Whitehead et al., 2011; Liljestrand, K., 2017; Kaipia, Dukovska-Popovska and Loikkanen, 2013). Different types of retail stores i.e single local retailers, multiple retailers/ retail chains and butchers have a similar supply chain structure. However, because of their different business models certain players in the industry perform activities of different stages of the supply chain, the same applies to producers. These differences in the structure can be understood by studying the different players in the Industry. Producers like Danish crown (2020) perform all parts of the production viz. Slaughtering, cutting and packaging. They not only have their own range of consumer products but they also provide industrial cuts. Additionally they also produce and package private label

products for retailers. Producers like Scandi Standard (2020) perform all parts of production but only have their own range of consumer products such as Kronfågel. Retailers like ICA act as the wholesalers for all their stores owned and run by different individuals. Along with procuring products of different brands, they also sell their private label products (Icagruppen, 2020). Retail chains such as Hemköp and Willys owned by Axfood AB (2020) have Dagab as their wholesaler, which is also owned by Axfood. All their private label products are produced for Axfood by different producers and repackers. Retailers like Citygross (2020) for their private label products, have cutting and packaging facilities in their stores for beef and pork, which means they procure industrial cuts instead of prepackaged consumer size units. However, for poultry, they follow methods similar to other players.

Apart from the primary activities which allow for a safe and good quality product to be delivered, players in the supply chain perform other activities in order to function efficiently, increasing profits and decreasing losses. Producers not only process the animal but have to do so in a safe manner by following safety protocols such as HACCP. They are also responsible for estimation of shelf life of products (Liljestrand, K., 2017; Whitehead et al., 2011). Similarly, wholesalers not only handle procurement, warehousing and storage, they also plan for proper materials handling and transport planning along with the logistics partners. In order to ensure availability of the product to the end consumer one of the most important activities is demand forecasting. This is done by both retailers and the producers (Kaipia, Dukovska-Popovska and Loikkanen, 2013; Liljestrand, K., 2017). Retail stores create the demand forecast and place an order for the required products and quantities with their wholesaler. The wholesaler aggregates all the order information from different retail stores and places orders with the producers. The lead time for these orders in the fresh meat chain can be as short as 24 hours. This depends on the producers as some of them follow MTO (make to order) and some follow MTS (make to stock) strategies of production, there are however many who follow a combination of both the rules since they have to take into account the supply side of forecasting as well though the supply side variability is low. Using forecasts they start the production based on MTS but they later change the production status and the flow based on the MTO rule when the orders are received from the wholesalers (Kaipia, Dukovska-Popovska and Loikkanen, 2013). Due to demand variability and based on the processes they follow some producers even freeze the meat after production as a last resort of avoiding wastage, this however increases costs and reduces profit margins (Whitehead et al., 2011). Additionally, coordination mechanisms such as trade agreements, information sharing and joint decision making are followed in order to coordinate these activities between players of the FSC (Liljestrand, K., 2017).

3.3 Fresh Meat Waste

Food waste has been defined using different expressions in literature, being a diverse product group addressing with a simple definition that fits all is complicated. These expressions used are sometimes overlapping and sometimes not. Therefore, it is important to be sure what it is defined as in each source that is referred to, especially while combining quantifications of food waste (Eriksson, 2015). Edible and inedible parts of food that have been discarded from the FSC and recovered in destinations other than animal feed and chemical processing such as landfills, composting, incineration, and sewer is defined as food waste (Tostivint et al., 2016). Additionally causes of waste such as animal death, production waste such as trimming spillage which the FAO (2011) defines as food loss are not of relevance to this research as only causes of waste that are related to the durability dates of food are taken into account. However, a distinction is made on the type of food waste in line with (Quested, Ingle and Parry, 2013) and Swedish EPA (2014). Unavoidable waste is the part of food waste that was not edible under normal circumstances such as meat bones and vegetable peels. Avoidable waste is the part which is considered edible by the vast majority of people at some point in time before it has been thrown away. Possibly avoidable waste is the part which is edible prior to disposal but eaten only by some people such as potato skins. With these definitions in place we now explore the causes of food waste.

3.3.1 Causes of Food Waste

Food waste occurs at all parts of the FSC, the quantities of waste, the burden on the environment and the reasons for waste are different for each part of the chain. Another fact presented in literature is the difference between countries in relation to waste. While waste is more prominent in the processing and production stages in developing countries due to lack of proper storage and skill, the majority of the waste in developed nations such as the ones in the EU occurs in the latter stages of the chain such as retail and end consumption. The same trend is seen in the FFSC(Göransson and Nilsson, 2013; Mena et al., 2014; Chen et al., 2020). Waste also occurs at the interfaces of the actors of the FSC, the causes of waste are interdependent and some even enhance the risk of food waste (Liljestrand, 2017). An important learning is that the stage where waste is accounted for need not be the same stage that has caused the waste. For example, when products of very short shelf life are delivered to the retail stores, waste occurs as they do not have enough time to sell the products. While the food is thrown away and accounted for at the store, the cause is actually at the wholesaler or producer stage where most of the products shelf life has been passed due to some reasons. While this is a case of waste being forwarded in the chain, a backward effect such as the bullwhip effect where implications of changing consumer behaviour in store increase in amplitude in the previous stages leading to waste (Eriksson, 2015; Mena, Adenso-Diaz and Yurt, 2011). This is ofcourse coupled with the lack of timely

communication between the FSC actors, restating that food waste causes are not independent of each other. Not only are the causes interdependent but they overlap between the stages of the chain based on the view point. For example, product returns are seen as a cause of waste in the retail stage due to inflexible standards. The same product returns can be seen as a waste in the production stage due to low operational performance leading to substandard product (Mena et al., 2014; Teller et al., 2018; Jevinger, Båth and Göransson, 2014). Allowing for this possible overlap, the causes of waste in each part of the FFSC are now presented stage by stage.

Production:

The production stage which includes slaughter, cutting and packaging has many areas where waste can occur. The waste occurring due to poor manufacturing processes is a food loss issue and not relevant to this research as mentioned before. However, there are other poor practices that also lead to waste in different ways. The natural constraint of demand variation and leading inaccurate forecasting is a major cause of waste. Producers tend to follow a make to stock principle in order to maintain high service levels leading to overproduction (Göransson and Nilsson, 2013). While this can be solved with collaborative forecasting most supply chain players lack the timely communication and most supply chains are not vertically integrated. This lack of coordination and transparent communication is seen as a cause of food waste (Mena et al., 2014; Teller et al., 2018). Another major reason attributed to waste is the product returns, this occurs due to many reasons of non compliance with merchandising standards. Products with wrong packaging and printed labelling, products with shorter shelf life than agreed upon, deviations from delivery times product damage in transit and deviation from standard temperature during delivery are the main reasons for returns (Göransson and Nilsson, 2013; Jevinger, Båth and Göransson, 2014). In terms of improper packaging and wrong labelling even if the deviation from requirement is detected in the production stage, the valuable shelf life of the product is wasted in rework, this of course also leads to creation of packaging waste (Mena et al., 2014).

Wholesale and Retail:

Demand variation, a natural constraint of fresh food products, is a major root cause of meat wastage. This variation occurs due to weather and seasonality which change the consumption pattern, promotions and marketing campaigns which make it harder to predict demand (Mena et al., 2014). Promotions and campaigns created to attract consumers cause demand variation not only in the particular product but in the whole range as the promoted product substitutes or cannibalises the sales of other similar products (Göransson and Nilsson, 2013; Eriksson, 2015). This causes unpredictable variation in demand even when planned for the whole range. This variation in demand coupled with inaccurate or poor forecasting practices cause the supply demand misalignment and the short shelf life of the product augments waste creation (Teller et al., 2018). Other than causing demand variation, promotions are also attributed to causing waste

at the end consumer level as certain promotions push larger packages and purchase of more units, which coupled with the short shelf life leads to waste (Mena, Adenso-Diaz and Yurt, 2011).

Poorly functioning supply chains is another root cause of waste creation. Cold chain mismanagement or breakage in cold chain due to equipment failure and temperature fluctuations lead to decrease in shelf life of the product. This mainly occurs at the interfaces of the actors such as the wholesale and logistics, supplier and logistics where lack of knowledge and communication leads to improper handling (Göransson and Nilsson, 2013). A simple example for this can be products left on the docking bay for too long causing temperature differences leading to shorter shelf life. In case of static durability dates this becomes a food safety issue as well. This along with the retail quality standards cause the product returns as mentioned in the producer stage. Additionally, measurement of temperature at the retail store can possibly reject a large number of packages based on a small sample (Jevinger, Båth and Göransson, 2014). In store mismanagement by employees also causes waste. Damage to products/ packaging due to mishandling both due to customers and employees causes waste. Additionally, lack of skilled employees leads to poor stock rotation in turn leading to waste. Though training is given a high turnover rate of employees and hiring unskilled workers during holiday seasons due to necessity makes it a challenge to keep every employee trained and motivated (Teller et al., 2018).

Certain practices in the retail industry which are done to maintain customer satisfaction and be competitive also cause food waste. Increasing width and depth of assortments make it harder to manage multiple short shelf life products and create more complexity for forecasting. Requirements such as hundred percent on shelf availability leads to overstocking of products with fear over losing sales, while neglecting or considering the waste associated with it to be a cost of proving the all time availability (Teller et al., 2018; Mena, Adenso-Diaz and Yurt, 2011). In these cases relative waste is more prominent in products of low turnover however in absolute terms higher turnover products have higher amounts of waste (Eriksson, 2015). Similarly, over ordering to keep shelfs full is a reason for wastage. Retailers also have very strict standards on the suppliers such as the amount of shelf life, deviation from which will lead to returns. Even if retailers buy such products for a discounted price from the supplier, the delivered product too close to the durability date makes it challenging to sell them within time leading to waste. Additionally, retailer's metric of serviceability pushes suppliers to maintain a buffer and overproduce in order to keep the serviceability levels high, which in case of short-life products is a risk factor of waste (Mena, Adenso-Diaz and Yurt, 2011).

Products in stores are mainly discarded due to passing of the durability date, damage to product and visual appearance such as discoloration when delivered. While these are the reasons the above presented are the root causes leading to the passing of the durability dates (Eriksson,

2015). Though the focal freshness sensor does not tackle the root causes, it intends to utilise the shelf life of a product to its full extent and avoid premature discarding of products. The real value such a sensor can provide is discussed in more detail in the results section of this report.

End Consumer:
The reasons for food waste at the end consumer stage are two fold. One part is attributed to their behaviour in store (this can be also considered a part of the retail stage). The other part is attributed to their consumption behaviour at home. Both of these behaviours are related more to psychographic factors rather than sociodemographics. Consumers tend to buy products with the longest shelf life even though they would consume it very soon, this leads to poor stock rotation in stores leading to waste. The reason for this could be that consumers feel products close to durability date are not fresh (Mena et al., 2014; Aschemann-Witzel et al., 2015). Purchase of larger quantities than required and not being able to consume them before spoilage, over estimation of meal sizes in households also leads to waste. Strict adherence to the durability dates given on the product packaging is another major cause of edible waste at the consumption stage (Filho and Kovaleva, 2015; Quested, Ingle and Parry, 2013; Jevinger, Båth and Göransson, 2014; Fanelli and Di Florio, 2016). This occurs both due to misinterpretation of use by/ best before dates and due to lack of experience in knowing the quality of food, leading to fear of food safety. Additionally, lack of storage knowledge also leads to shorter shelf life of the product, this mainly includes the temperature deviation during transport from store to home and higher temperatures in the refrigerator (Marklinder and Eriksson, 2015).

Extant literature which presents these multiple causes of food waste also mentions the need for all parts of the supply chain to work together to tackle the problem (Teller et al., 2018). Starting from understanding the hidden costs of the problem and its impact on the bottom line are the catalysts for resolving this problem (Mena, Adenso-Diaz and Yurt, 2011). A change in top management's view on these issues from waste being a necessary part of the value delivery is needed (Eriksson, 2015). Additionally, focusing on solutions for internal waste reduction i.e each stage by themselves can lead to suboptimal results as that could push the problem from one stage to the other (Mena et al., 2014). While there are many causes of waste, the most relevant ones to this research are the ones related to durability dates which mainly occur at the retail and end consumer stages. The magnitude of fresh meat wasted due to passing of expiry dates both in retail and end consumer stages is explored next.

3.3.2 Magnitude of Fresh Meat Waste

While there is a rise in interest regarding food waste, the quantification and accounting of food waste still lacks studies which have a systematic approach mainly to account for waste in a disaggregated manner in the different supply chain stages and different food groups. The extant

accounting and the frameworks for the said accounting is at an early stage and challenging. There is a variation in the methods and definitions used in the studies which makes comparison and verification of data difficult (Caldeira et al., 2019). The availability of data in terms of geographical regions, supply chain stage and food commodity varies strongly (Bräutigam, Jörissen and Priefer, 2014). This can be clearly understood based on the critical review on food waste quantification by Xue et al (2017). Out of the 202 reviewed publications until 2015, the majority mention or focus on the USA and the UK, with Sweden being the third largest mentioned country. Similarly Households and retail stages are the most covered in these publications. Though Sweden is the third highest mentioned country, most studies only provide an aggregated food waste quantification, very few provide a food category level data. While 8 of these publications mention meat waste in Sweden only 5 of the publications have provided a certain amount of detailed data on meat wastage. Xue et al (2017) also presents the different methods used to quantify food waste, both direct and indirect methods of quantification have been used with only 20% of the publications using a direct method. Direct methods include first hand collection of data by weighing waste at different stages sometimes including a compositional analysis. Furthermore data is collected through surveys, dairies and records from retail organisations. In case of indirect methods studies use mathematical models, proxy data and literature data to quantify waste. Over 40% of the studies use literature data i.e data collected in other studies to make their estimations, this also indicates that some publications have been cited extensively to use the secondary data and any discrepancies in that data would reflect in the estimates. Both these methods have advantages and disadvantages mainly in terms of accuracy, representativeness, time and cost. While direct methods are the most accurate they are expensive and time consuming and cannot be representative of the whole country or stage of FSC. Indirect methods make up for representativeness but in turn have lower accuracy. Additionally, Xue et al (2017) presents that most of the data on food waste is from the years between 2008 and 2014, mainly in 2011 and 2012.

Bräutigam, Jörissen and Priefer (2014) uses mathematical models and secondary data from various publications and estimates the food waste at EU level to be 142.748 million tonnes during the year 2006. Similarly, Caldeira et al (2019) use secondary data to conduct a mass flow analysis and estimate the food waste at a EU level to be 128 million tonnes during the year 2011. This difference in the estimates is due to the different methods used and possible decrease in waste over the years. Caldeira et al (2019) also provides data on food category level, accounting for 14.2 million tonnes of meat from the 61.7 million tonnes of meat that entered the FSC. While 23% of total waste was in the meat category, 1.7% of the total waste occurred at retail and distribution and 7.3% at households. However, at an aggregate level 5% of the total waste occurs at retail and distribution, the highest amount occurs at the consumption stage which is 46%. Noting that these numbers include all types of meat products wasted due to different causes mentioned previously, the retail and consumption stages are now explored in detail.

Meat Waste in Swedish Retail:

Swedish EPA (2014) estimates that out of the 1211000 tonnes of food waste in Sweden in the year 2012, 7 kg per capita or 70000 tonnes was retail food waste out of which 63000 tonnes is unnecessary or avoidable waste i.e if handled correctly or eaten by the use by date could have been saved. Furthermore, they estimate that 29% of the retail waste was meat wastage. While these estimates are based on data calculated using a waste factor of food waste per employee which is in turn based on the compositional analysis of waste at centres collecting supermarket waste the other major study Eriksson (2015) provides first hand data based on the research involving six Swedish supermarkets during the years 2010 to 2014. As the studied six stores are of the same format and retail chain in the same region, this data provides insights into the detailed level of meat wastage in the retail stage but is not representative of the national data, use of such data from limited case studies to calculate national data can give variation in results. Additionally, Eriksson (2015) divides waste into prestore and instore waste. Prestore waste includes waste due to rejections by the retailers, Instore waste includes waste due to unsellables i.e damaged products, visual discoloration and products with passed due date. The meat department in the studied stores sell products of beef, pork, chicken, lamb, game meat, grilled chicken, raw sausages and some frozen meat. The meat department in all six stores combined from the year 2010 to 2014 contributed 88 tonnes of waste by weight or 3.67% of the total wasted food mass in the six stores, which is worth 5.3 million SEK or 11.78% cost of waste and 1200 tonnes CO_2 equivalent or 30.78% of the carbon footprint due to waste. Majority of the waste in the meat department is instore waste, prestore waste accounts to a very small amount which ranges from 0.004 tonnes to 0.69 tonnes in the top 14 categories of waste in the meat department. Additionally, it can be seen that the top 20 products contribute to 81% of the waste in the department both by weight and CO_2 equivalent. While pork chops contribute to the highest amount of waste by weight at 7.8 tonnes, minced meat is the highest wasted in terms of carbon footprint at 200 tonnes CO_2 equivalent generated from the 7.1 tonnes. This data is accurate and reliable as the meat wasted is recorded using the EAN barcode on the products and the influence of fluctuating amounts of waste over time is reduced as the study is conducted over five years. This data represents the average waste of the particular format i.e discount stores; however, there could be limitations as the studied six stores were chosen by the retail chain and there is a small possibility of unrecorded waste (Scholz, Eriksson and Strid, 2015). Additionally, this data does not provide insights into the quantification of waste in relation to the different causes of waste. While such detailed data was not available in Sweden, a study by WRAP in the UK estimates that 38% of wastage in the meat department is from passing of due dates based on data from three retail supermarkets (Anthesis et al., 2018). Studies in other nordic countries also mention passed expiration dates to be the main cause of waste in retail, however they do not provide any quantification on such a level for meat (Stenmarck et al, 2011).

Meat Waste at the Consumption Stage:

Swedish EPA (2014) estimates that in the year 2012, 771000 tonnes which is 81 kg per person of food waste from households has occurred, out of this 270000 tonnes which is 28 kg per person is estimated to be unnecessary waste. In relation to household food waste in Sweden Schott and Andersson (2015) present the waste composition analysis of 486 households from 2008, where meat waste was composed of unopened and opened packages, prepared, half eaten food. They have found that 10% of the total waste from meat was avoidable waste. While opened packaging was 42.5% of total avoidable waste, 6.6% of avoidable waste was unopened packaging collected from residual waste. In relevance to this research meat that is thrown out due to a passed due date can be part of both opened and unopened packages. However, knowing the exact amounts is highly challenging as mouldy or spoiled meat is also a part of this waste. The limitation from Schott and Andersson (2015) is that they do not provide the numbers on household food waste in Sweden; rather they estimate it to be 38% of the solid household waste. Even the Swedish EPA (2014) does not provide detailed data at the food commodity level. However, data at a more detailed level is available for the UK, while this data might not represent Sweden it gives an idea about the composition of household food waste in a similar country. Compared to Sweden EPA (2014) data the UK household food waste is estimated to be 110 kg per person. While this means Swedish food waste is lower, we need to bear in mind that it is possible that Swedish data has not accounted for all the waste that the UK study has accounted for.

Quested, Ingle and Parry (2013) have reported the household food waste data in the UK for the year 2012, they have used a combination of waste composition analysis, surveys and waste dairies to make these estimates. Both fresh and processed pork, beef, lamb and poultry (mainly chicken but include turkey and duck) were the categories of meat in the estimated data. Avoidable food waste that has occurred due to not consuming in time i.e meat that has passed the due date or has gone off or is mouldy is estimated in this study. It is to be noted that the food thrown away with more than 40% of meat on it was considered avoidable though it contains unavoidable content such as bones. A total of 280,000 tonnes of poultry, 140,000 tonnes of pork, 56,000 tonnes of beef by weight are estimated to be wasted out of which 110,000 of poultry, 99,000 tonnes of pork and 34,000 tonnes of beef was avoidable waste. This avoidable waste is estimated to be of the value £660 million, £640 million and £260 million for poultry, pork and beef respectively. It is observed that relative unavoidable poultry meat waste is a larger part compared to pork and beef as most poultry products are sold as carcass meat resulting in approximately 50% of waste being unavoidable. When looked into the reason for waste it can be seen that avoidable food waste of 31,000 tonnes/ £190 million of poultry, 47,000 tonnes/ £300 million of pork and 10,000 tonnes/ £76 million of beef was thrown away as it was not used in time. The study also provides data on a more detailed level for poultry and pork. Total waste by weight of fresh poultry(carcass meat) is estimated to be 240,000 tonnes of which 78,000 tonnes/ £390 million is avoidable. Total waste by weight of fresh pork (carcass meat) is estimated to be

46,000 tonnes of which 21,000 tonnes/ £120 million is avoidable. While these details are not provided for beef, minced meat as a product is also not mentioned as it would not be possible to identify it in the waste composition. While these numbers would be different for Sweden due to differences in consumption of meat types, the part of waste that is caused due to not using in time could provide some insight into the human behaviour of wasting. 28.18%, 47.47% and 29.4% of avoidable waste of poultry, pork and beef was wasted respectively due to not using the product in time. In comparison to other European countries it can be seen from a similar study in the Netherlands in 2010 that 27% of the fresh meat wastage is due to expiration dates. Additionally, it was reported that the meat waste was only 2.5 kg per capita per year out of the total avoidable food waste of 44 kg per capita per year (Soethoudt et al., 2013).

3.4 Food Legislation

Spoilage sensors such as the focal product are defined as per *Article 2* of the regulation EC 1935/2004 as intelligent food contact materials, those which monitor the condition of packaged food or their environment. This regulation provides provisions for food contact material which are not inert like traditional food contact materials, substances which are not already included in the list provided by the authority can obtain authorisation as per *Article 8* of the same regulation. Additionally, in the light of food safety these materials need to be manufactured in agreement with good manufacturing practices. This can be achieved by following protocols such as hazard analysis and critical control point, HACCP similarly to other food supply chain actors (Mortimore and Wallace, 2013). Further requirements related to the labelling of the material and the information that needs to be provided for consumers are mentioned in the above stated regulation. These legislations are directly related to the product and are needed to be accounted for in the product development process. However, there are other legislations of relevance to the business models, which are to do with the date labelling and sale of packaged food products.

As per the food information to consumers legislation, *Article 9* of the regulation EU 1169/2011 all food products with certain exceptions must provide certain mandatory information of the product to the consumer, one of which is related to its durability. The regulation allows the use of durability dates of two types (1) Best before date (BBD) or (2) Use by date. While a use by date is to do with food safety, best before dates are related to the food quality. The following are the definitions of these two as stated in the legislation as per *Article 2, Article 24 and Annex X* .

"Date of minimum durability of a food' means the date until which the food retains its specific properties when properly stored", "The date shall be preceded by the words: 'Best before …' when the date includes an indication of the day, 'Best before end …' in other cases"

"In the case of foods which, from a microbiological point of view, are highly perishable and are therefore likely after a short period to constitute an immediate danger to human health, the date of minimum durability shall be replaced by the 'use by' date"

The responsibility of conducting the studies such as microbiological tests to estimate shelf life falls on the manufacturer of the food product as per *Article 3* of the regulation EC 2073/2005. Which means the responsibility of choosing the type of date to use for a particular food product is relied on the food manufacturer. Furthermore, in cases where food is repacked or reprocessed by other food business operators the responsibility of safety and labelling lies with the packer of the food (Livsmedelsverket, 2020(a); Soethoudt et al., 2013). Additionally, *Article 3* of the regulation EC 2073/2005 requires that the safety criteria be met under foreseeable conditions, which means a safety margin be applied on the estimated shelf life (FSAI, 2019). While there are no regulations specific to the selection of the type of durability date several government and non-government bodies of the different EU states have provided guidance with decision trees and recommendations for the responsible parties (FSAI, 2019; FDF, 2017; Livsmedelsverket, 2020(b); WRAP, 2017). There is however an exception in the case of fresh poultry meat, as per *Article 5* of the regulation EC 543/2008 which lays down the marketing standards of poultry meat the durability date for such products needs to be a use by date.

The other relevant legislation is regarding the sale of food in relation to the durability date used. As per the regulation EU 1169/2011, the sale and redistribution of best before products is allowed as these only indicate quality and not safety. However, use by date products are considered a safety risk and therefore strictly not allowed to be sold or redistributed after the mentioned date has passed. While the food information to consumers legislation regarding durability dates is followed in all member states of the EU and EEA, the interpretation of law and the sale of food after durability date is not exactly the same (Wageningen Economic Research, 2020). While none of the EU states allow for sale or redistribution of the food after use by date, certain states do not allow the sale or redistribution of foods which have passed best before dates as well. The law in Sweden however allows for the sale and donation of food after the best before date has passed but the seller needs to ensure the safety of the food i.e the liability of food safety transfers from producer to retailer at this stage (Livsmedelsverket, 2020(a)). Though the legislation allows the lack of tools to easily assess the quality of food at the retailers and the misconception of health hazards are the main reasons for retailers not selling such food (Wageningen Economic Research, 2020). Additionally, both best before and use by products are allowed to be frozen for preservation and sold, while best before date products can be frozen after the date has passed, use by date products can only be sold if frozen before the date. Best before date products can also be upcycled (used as raw material for other products) later in time. Such freezing and sale of products should meet the legislation related to frozen foods (Livsmedelsverket, 2020(a)). Heterogeneity regarding these issues exists across the EU and the

Nordics. Soethoudt et al. (2013) and Hanne et al. (2015) provide information regarding the practised legislation in all EU member states and Nordics respectively. The latvian legislation which is of relevance to this research does not allow for the sale or redistribution of food after the durability date has passed. However, the ministry of agriculture, Latvia as of 2018 is working to amend this law in the light of food waste.

Presently there is no legislation regarding the sensors as a food information to consumers. This allows for the use of sensors alongside the traditional date label. Sensor producers such as Keep-it (2020) are evidence of the same. This however might be challenged in the future as the data provided by these sensors can be interpreted as information and the presence of both traditional and sensor information can confuse the consumer. While this is the case in the EU, bigger markets such as the US still do not have a standard legislation regarding food date labelling (USDA, 2019; ReFED, 2020). The company therefore has to take in account these differences in markets and make changes to the business model as required.

3.5 Dynamic Shelf Life Market

Intelligent packaging technologies are emerging to provide different values to the food industry such as food safety and traceability, these technologies are now also being provided to determine the shelf life of a product in real time acting as a potential solution for food waste, mainly in the retail and consumer stages (Sohail, Sun and Zhu, 2018; Poyatos-Racionero et al., 2018). There are several of these technologies working with different principles, some of which have been tried to commercialise and failed and some of which have been in the process of commercialization since the last few years (Poyatos-Racionero et al., 2018; Mustafa and Andreescu, 2018.). Time temperature indicators (TTIs), freshness indicators, dynamic shelf life (DSL) prediction systems which provide similar functions as the Innoscentia analogue sensor are discussed below. Additionally, emerging dynamic pricing systems which also provides a potential solution for food waste in retail is discussed.

TTIs:
The basic idea of these products is based on the fact that temperature changes induce microbiological growth and its acceleration leading to food spoilage. However, the active ingredient and the working principle which allows us to measure temperature change over time are different. Varying from physical reactions such as the 3M monitor mark where the substance melts at a set temperature, chemical reactions such as Keep-it, Evigence (2020) and Insignia and enzymatic reactions such as Vitsab and Topcryo by Cryolog. All these indicators still provide a visual indication of change (Kerry and Butler, 2008; Poyatos-Racionero et al., 2018). While some TTI indicators such as Timestrip (2020) provide the history of temperature of the product, others such as Vitsab (2020), Evigence sensors are more user friendly providing an indication of

freshness. Products like Keep-it (2020) are designed to show the number of days left. The value offered and the target segments of these products are clearly different. Products such as Cryolog (2020)and Timestripe are quality monitoring solutions for cold chains, where the TTIs are placed on secondary and tertiary packaging. Keep it and Vitsab labels on the other hand are placed on primary packages i.e each unit of food and aim to indicate freshness to the end consumer in real time. These companies have also provided successful use cases on their websites with Keepit targeting the retail fresh meat market and Vitsab and Timestripe in the seafood market. Companies like Evigence (2020) and Insignia technologies (2020) offer both types of labels. Evigence additionally provides a consumer mobile application to further enhance the use of the product. These products however have not been able to penetrate the food retail market with exception of Keep-it which is commercialised only recently. They have been limited to the application of ensuring cold chain integrity mainly by producers (Göransson and Nilsson, 2013). Additionally though the technology has become more reliable over the years it is still an estimation based on temperature. TTIs have to replicate the behaviour of food, knowing the initial microbiological status of food is a challenge which indicates that TTIs will have some safety margins and might not provide better estimations compared to that of direct measurement by freshness indicators (Kerry and Butler, 2008).

Freshness sensors:

Technologies such as Innoscentia´s which estimate the food product's microbiological state by directly measuring the metabolites which are products of biochemical changes/microbial deterioration come under this section of intelligent packaging. The working principle of such technologies can differ by the metabolite being measured such as organic acids, carbon dioxide, biogenic amines, glucose, volatile nitrogen compounds, sulphur derivatives and more. While there have been many attempts to develop such technologies, successful commercialization of them is still rare. Lot of these technologies are still in the development phase and some of them such as Freshtag detecting volatile amines, Toxin guard and Food sentinel system to detect pathogens using antibodies which were commercial products are now discontinued (Poyatos-Racionero et al., 2018; Fang et al., 2017; Sohail, Sun and Zhu, 2018). The reasons for the failure of these products are unclear. Recent technology development by researchers at the Imperial college london has given a low cost (lab prototypes made at two US cents each) freshness sensor based on which the company Blakbear was incorporated. They use a paper based electrical gas sensor or PEGS for short to detect metabolites such as ammonia and trimethylamine which are components of total-volatile-basic-nitrogen (TVB-N) produced when meat and fish products decay (Barandun et al., 2019; Imperial College London, 2019). These sensors are connected to NFC to read the freshness indication. Blakbear (2020) provides digital labels and aims to provide value to producers, retailers and consumers. Such digital technology is definitely considered competition as they not only provide the same product information but allow it to be shared digitally, which is an advantage in the supply chains. This also provides new

possibilities such as product marketing to consumers during purchase (TPG Rewards, 2020). However in retail stores this will require consumers to use their phones to read the information, whereas the analogue sensor might be user friendly in that aspect.

DSL prediction systems:

These systems use sensors to collect and log data across the supply chain, mainly the temperature which is used to predict shelf life of the product, this predicted life information can be accessed through the passive or active RFID chips present on the package both by end consumers and other stakeholders. Researchers have presented the required architecture for such systems and have done field studies to understand its potential. These systems will require sensors on the package, readers to obtain the data from the loggers and sensors, cloud systems to store the data (Jevinger and Davidsson, 2016; Jevinger, Göransson and Båth, 2014). Companies such as TempTrip (2020) provide traceability across the supply chain using wireless loggers and provide reports to supply chain actors; however, they do not target end consumers nor do they predict DSL. Companies such as Tsenso (2020) have now developed FreshIndex (2020) as a product for the whole supply chain. Consumers can use a mobile application to understand freshness of the food, retailers and producers can use the Freshindex system to access analytics and explore value from data. The main barriers for the adoption of these technologies are the larger investment costs involved to build the required architecture and the lack of collaboration between supply chain actors (Raab, Petersen and Kreyenschmidt, 2011; Göransson and Nilsson, 2013). Additional advantage of these technologies is the economies of scope, i.e the same system architecture can be used to monitor different products however, for that the behaviour of the products in non-isothermal conditions needs to be fully understood which is an additional challenge (Raab, Petersen and Kreyenschmidt, 2011). These systems can be perceived as both competition and potential collaborators. A clear collaboration opportunity arises in the case where the temperature indicators are substituted for freshness indicators.

Dynamic pricing systems:

Discounting, in recent years, has been used as a strategy to sell products with lower shelf life and inturn reduce waste. Companies such as Wasteless (2020) use AI and data analytics to provide dynamic pricing based on the product quality (using traditional durability labels), stock data and customer behaviour. This is basically a smarter way to attain optimal timing and magnitude of discount. These systems also provide value to suppliers through data analytics for evaluating performance and forecasting. Similarly, Whywaste (2020) provides data driven solutions to manage stock based on expiry dates to suggest markdowns and also provides a platform to resell. Such systems at the retail level aim to potentially reduce waste just like the focal technology. These products can be seen as a competitor and as a collaborator. Similar to DSL systems these products provide economies of scope i.e ability to use it on several food products rather than just meat and taking into account limited budgets for such solutions retailers might tend to choose

only one of the technologies. However, a combination of both technologies is much beneficial for retailers, as this will ensure better profits when products of least shelf life are discounted i.e more profit when DSL is used instead of fixed shelf life (Buisman, Haijema and Bloemhof-Ruwaard, 2019). Additionally, the focal technology allows the consumers to be sure of the safety and potentially reduces food waste at the consumer. With these two reasons in mind industry players might be keen to invest in both technologies.

4. Methodology

4.1 Research Setting

Innoscentia AB is a foodtech startup with its business unit currently based in Stockholm and R&D unit in India. Additionally, its board of directors are based in Lund, along with the label manufacturing partners. Aiming to commercialise an invention from Lund university the founders of the company set out to tackle the food waste problem. After gaining an initial interest from supply chain players, this company was established using soft funding in the year 2015. Since then the company had been in a deep dive R&D phase to develop the invention for commercial use. Few years later the decision was made to explore the market and path to commercialization of the product which led to the opening of the Stockholm office. As a part of the process this thesis was commissioned to learn about the market and provide multiple business model ideas that the company can later test and develop further. As of today the company is approaching the production of working prototypes of the product and aim to perform pilot tests in the near future. The protein source for which the sensor is currently being tested is chicken, however with further development/ calibration it can be used for other meat such as beef, pork and fish.

The Innoscentia analogue sensor is a freshness sensor which provides a visual change in relation to the change in the meat product. The chemical formulation present on the label reacts to the gases produced when meat deteriorates over time due to microbial growth. The working principle requires the printed side of the label to be in contact with the atmosphere of the meat package. Though this means the current product cannot work with vacuum packages, the company representatives feel they can develop a sensor for the same. Additionally, in relation to the product development the exact industrial design of the label and the way it will be integrated with the packaging process of meat is not established. Furthermore, the company is in parallel developing a digital sensor with a similar working principle where data on quality of the meat can be obtained through an RFID/NFC chip attached to the circuit.

4.2 Research Design

The aim of the thesis to design business models was based on the initial discussion with Innoscentia and their current needs. Furthermore, discussion with the thesis supervisor led to the choice of exploring multiple business models which is also based on the understanding that business models are established over time and new experimentation in real time. As such experimentation which is longitudinal in nature is out of the scope of this research presenting multiple business models which can be the basis for further experimentation by the company is

chosen. A literature study was conducted to establish the frame of reference which led to the use of BMC and the need to study the internal and external environment of the firm. The internal environment is flexible as the firm is at a startup stage therefore focus was on studying the external environment mainly to understand the value of the product being developed, characteristics of the supply chain stakeholders which would then be used to design the nine elements of the business model. The external environment was studied using different methods, firstly a literature study to understand the product and the problem it is trying to solve were conducted. Simultaneously, literature study on the supply chain stakeholders and the semi structured interviews were conducted. The results from this literature study are presented in the industry context chapter. The literature study is conducted to discover patterns and increase understanding which is abductive research (Alvesson and Sköldberg, 2008). Findings from the literature and interviews resulted in one, establishing that changes in the processes followed by stakeholders in relation to previous research were minimal. Two, there was a disagreement on the magnitude of waste due to traditional due dates. As this would have an effect on the value proposition of the product it was decided at this stage to conduct a case study analysis to evaluate the existing understanding of value. It is to be noted that conducting a quantitative analysis in order to increase the quality of study by triangulation i.e using different sources of data was considered from the initial stages. However, the unit of analysis for the quantitative part was established at this stage in the research. The choice of a case study was made as it provides the ability to gain deep insights when there is a lack of previous research, it is suitable for preliminary investigations and as a foundation for future research (Saunders et al., 2016). As it is a suitable method to find answers to specific how and why questions, the single case study approach is chosen (Yin, 2009). Additionally, The choice of performing the case study on the retail stage of the supply chain arises from the literature study conducted. It was plausible at this stage that retailers not only gain the highest value comparatively out of this sensor but also are the dominant actors in the supply chain. The case study is performed in a Baltic retail chain and consists of both qualitative and quantitative methods. The qualitative part of the case study was conducted using semi structured interviews with a goal of understanding the functions in the retail stage in more detail. In depth interviews allow for gaining deep insights from “experts” on the issue being studied. The open ended and semi structured style allows for interviewees to express all their views and provides the opportunity to rephrase questions and explain terminology to obtain responses unlike fixed surveys and questionnaires. These one on one interviews also provide the space to discuss sensitive topics where the interviewee can express opinions without feeling judged (Guest, Namey and Mitchell, 2013). The quantitative part was done by the procurement of records on food waste with an aim to evaluate due dates as a cause of waste. After the analysis of the data from the case study all the results obtained so far have been used to build the business models. The content and the architecture of the model could be changed to give rise to multiple business models. Firstly, the value proposition of the sensor to different stakeholders was established based on the different use cases i.e best before date and

use by date product. After this different ways of value creation and delivery were established which also affect the value proposition partially. The cost structure and revenue structure were a result of the above choices and not the choices themselves. The business model design was another qualitative approach used in this research which is based solely on the authors skills of creativity and analytical thinking. These business models were then evaluated by the author qualitatively in order to let the reader understand the implications of the different choices made in the business models. The choice of the comparative metrics chosen is based on the frame of reference conducted initially which shows that good models need to have a configurational fit and fulfil both short term and long term needs. Short term needs of being profitable, long term needs of being scalable, rigid enough to prevent imitation and flexible enough in times of change.

4.3 Data Collection and Analysis

Data for this research was collected in three forms, literature review, semi structured interviews and case company records. Literature study included journal articles, chapters of books, documents from governmental organisations and organisation websites. These were accessed through Google scholar, Google search, DiVA and EUR-lex. A total of 13 Semi structured interviews were conducted as a part of this research, the list of these interviewees is presented in Appendix I. An interview guide with open ended questions was prepared before every interview keeping with the interviewee in mind, these guides often evolved as the research progressed, both new and leading questions were asked during the interview. The interviewees were identified based on their work. Interviews were conducted in person, over the phone or through video conference applications. While most interviews were recorded, few were not due to technical and practical difficulties. All the recorded interviews were transcribed later for analysis. However, unrecorded interviews were done with additional care by taking brief notes during the interview and making elaborate notes right after the interview to avoid missing critical information. In addition to the mentioned 13 interviews, meetings with the CEO and other employees of Innoscentia were conducted initially to learn about the current status of the company and the product. The case company data is obtained from the company records. This is data that is recorded in retail stores through the established daily working procedures. It can be accessed centrally through enterprise systems used by the case company. Data was collected by first sending a data request containing multiple questions to the company relating to quantities of food waste, causes of waste which can be seen in Appendix II. After a representative was then assigned from the supply chain department to extract the data from records, a first meeting took place where the employee had gone through the presented questions and clarifications were provided by the author to avoid misunderstandings. Weeks later another meeting with the representative was conducted where raw data extracted from the case company's enterprise systems was presented, the definitions, units, abbreviations used and the structure of the report

was explained by the employee. Data related to certain questions was either unavailable or deemed critical and not allowed to be shared at this stage. Later an excel file with the requested data was received from the company. The choice of the case company, a top Baltic retail chain, was mainly due to practical reasons. Retail chains in Sweden were contacted for such data, however the requests were declined or either left unreplied. The possible reasons for this are mainly the consideration of food waste data to be very critical, heavy work loads due to the Covid-19 pandemic.

All the interview transcriptions were read more than twice before the results were written. All the interviews were coded to categorise answers mainly into answers related to food waste causes, food waste magnitude, ways of working, current approaches to waste and perceptions on the sensor as a solution. It needs to be noted that not all interviewees were asked the same questions leading to some answers being coded as miscellaneous which were later categorised as relevant and not relevant to the study. The presentation of interview results also follows these categories. The interviews at the case company are analysed separately and presented within the case results, this is mainly to allow the reader to compare the results from interviews in Sweden to that of the Baltic interviews. The data from the records of the case company was analysed using Microsoft Excel 2013. Firstly, the data provided was structured in different ways to calculate the different results presented. Data on quantities of waste in different units and data on cause of waste were structured based on the product category to study the overall quantities. Another structure for the same data is made with reference to the store format. Next this structured data is recalculated for the period of one year. Calculations of total waste and relative waste of each product category are done. The part of waste due to the due dates is calculated and presented. Finally, using the sales figures an estimation of the number of sensor labels required in a year was made which was then used to calculate the plausible cost of the sensor label. Tables made from the structured data to be presented in the results section. The selection of the results presented is inspired from Eriksson (2015).

4.4 Validity and Reliability

The validity and the reliability of research is expressed based on the research design tests proposed by Yin (2009) for case studies. Using multiple sources of evidence including literature research increases the construct validity of the research. The external validity is obtained through the use of literature. As the data collection was done in a structured manner the reliability of the study is also high meaning if the study were to be done using the same methods again the results would be similar. The interview methods used also have high validity and reliability. The development of interview guides prior to the interviews in a structured way as presented in Guest, Namey and Mitchell (2013) increases the validity of the study. To be transparent the interview guides though discussed with the thesis supervisor were not externally reviewed

making the validity of the research depend on the author's knowledge. The interviewees selected were of high quality. They were able to articulate themselves in English and understand the researcher. All the interviewees are experienced in their field of work. This further increases the validity of the research. However, it is to be noted that there are limitations to the reliability of qualitative research, this is due to the subjectivity of the research. The conditions in which the interviews were done cannot always be replicated and the behaviour of the interviewees cannot be fully controlled. Additionally, due to the large amount of information handled in a qualitative study, choice of presented results is based on the research's perception of importance (Bell, Bryman and Harley, 2018). Additionally, the empirical results cannot be generalised as only a few interviews were conducted and as it is a single case study it need not depict the whole industry. Finally, the business models generated are based on the author's capabilities and hence a subjective result and the validity depends on the author's knowledge. Additionally, as the business models are a result of creative processes generally including many iterations those particular results do not have high reliability.

5. Results and Analysis

5.1 Learnings from interviews

Wholesaler:

Interview with the vice president of a wholesaler to the foodservice industry provided insights regarding the products viability in that market. The interview mentioned several trends in the food service industry. The restaurant orders were moving towards more convenience which means ordering smaller package sizes was an increasing trend, specific types of cuts of meat were ordered which are specific to the restaurants, this was an increasing practice to avoid cutting waste in kitchens and save working hours. In relation to the sale of meat beef and pork were most sold with an increasing trend in chicken sales. However, the interviewee mentioned that the green movement can be seen in Sweden and that the general turnover growth of meat in Sweden is stagnant. As their company also repackages meat, they take responsibility for the quality of the product delivered. However the due dates are established by the manufacturer, repacked product dates are related to them and regular quality audits are done to ensure the safety of the meat. While these learning are in line with the literature review, the interview did not see much use of the freshness sensor in this market as cooks and chefs in restaurants have long experience in using meat products and would not require a freshness sensor, rather they would use sensory analysis and in many instances use meat that has passed the BBD.

Food regulation:

The interview with the food regulatory affair specialist has provided answers very much in line with the literature review. While there are no specific regulations to curb food waste there are initiatives and state funded projects to measure and provide solutions to curb food waste. Over the years almost 95% of the regulations in Sweden are the same as that of the EU legislations. The choice of type of date label for a meat product falls on the producer except for chicken where only use by date is allowed. The food manufacturer is responsible for the safety of the food and liability shifts to retailers in cases of passed due dates and in case mistakes from the retailer side lead to the quality deterioration. The interviewee stated that legislative changes and amendments take a long time and this period could be 5 to 10 years. Additionally, the sensor's reliability was questioned and stressed as the most important factor to provide end consumers with valid data all the time. Furthermore, a remark regarding freezing meat during the last due date as an alternative to wasting was made.

Retail stores:

An experienced store manager, three experienced store employees and an innovation engagement lead from a retail chain were interviewed. While two of the experienced employees(1 and 3) worked in different stores of the same retail chain, the others worked in different retail chains. The routine retail store operations were similar in all the stores, employees removed products that have passed the due dates, damaged and unsellable products from shelves, along with this products are arranged in a first in first out order. Unsellable products were later recorded and discarded. The store manager, innovation lead and store employee 1 mentioned that when there is heavy workload and just one or two items to be recorded, employees tend to either throw the waste unrecorded or do not classify the waste clearly based on the preset reasons for waste. Furthermore, the innovation lead mentioned that this could lead to the waste data accuracy problems and that in their retail chain since the store managers are evaluated based on net profits there is a possibility of irregularities in recording waste. Store employee 1 also mentioned that department managers are given yearly bonuses based on net profits therefore they are very motivated to reduce food waste in general. However, it was mentioned that manipulating waste data by not recording it would be tough as it can always be traced by measuring the delivered mass to the store. In relation to store operations the ordering of food takes place automatically in all the stores. The store manager mentioned that he would make changes to the computer generated order which was based on past and recent historic data. These manual changes were made based on experience and weather patterns. All stores received delivery of fresh goods daily and made orders on a daily basis. Store employee 1 mentioned that due the structure of their organisation each store had their own terms with the parent organisation in relation to the cost of the products and delivery times.

Food waste was considered as unfavourable by all the interviewees however, the details regarding waste in each store differed. Store employee 1 and Store employee 2 mentioned that food is wasted in their stores and that due dates were a common reason for it. Though they could not provide an estimate of how much was wasted, store employee 2 mentioned throwing two to three packages per day on an average including all causes of waste. Store employee 3 however stated that fresh meat waste was uncommon in their store and the use of the why-waste system provides timely markdowns on the products and that most of these discounted products are sold. Similar to this the store manager mentioned that markdowns are used as a way to reduce the waste, they markdown products during their last day of printed shelf life or a day before that. It was also mentioned that one or two packages a day are marked down and many days all products are sold. The decision of mark down lies on the store manager, which is similar to store employees 1's comments where the department manager decides on the markdowns. The store manager further mentioned that around 25 kgs of meat products are wasted per week which is mainly due to damaged products and a small part of it due to expired due dates. Damage to products due to consumer behaviour of leaving products outside the refrigerators, puncturing the

packages and shuffling the products leading to disturbance of stock rotation were mentioned. While this number is quite low compared to the approximate 61 kg of meat waste as presented in the literature study, it needs to be noted that the store formats and the regional location of the store are different. Additionally the possibility of slight waste reduction due to better ordering and new measures compared to 2012 cannot be eliminated. Even store employees 2 mentioned the use of markdowns, however here the markdowns were done on a standard procedure and discounts were centrally controlled. In relation to markdowns store employee 1 mentioned that due to the possibility of cannibalising sales of similar products some managers see that it is better to waste than provide discounts. Additionally, as the area of the store is not in a price sensitive customer zone, their store does not provide many discounts on products. In terms of the products wasted the store manager mentioned that chicken and minced meat were the most wasted products in their stores, while the other interviewees did not have insight on this. Additionally, their store preferred having less tray packed goods and more vacuum packed goods and ordered accordingly. While more and more products are being vacuum packed, certain stores based on location and formats tend to prefer tray packs which have a head space i.e with an atmosphere unlike vacuum. This was observed in store employee 2's store. As these stores are low price stores it is possible that the reason for preference is the lower cost of tray sealed products. Additionally, it was observed that none of the chicken and minced meat products were vacuum packed.

In relation to the use of a spoilage sensor on fresh meat, store employees 2 and 3 did not comment much on its usefulness nor did they disregard its use. However, store employee 1 and the store manager mentioned that this would be a useful way for end consumers to know the quality of the product, they could not however mention strong uses for the retail stage though they thought it would be advantageous. The store manager and innovation lead further mentioned that it would be of interest to understand its implications on employees operations and that there would be a need for establishing standard practise to avoid confusion between the printed date label and the spoilage sensor. Finally, Commenting on the overall theme of sustainability the innovation lead stated that while their retail chain had established goals to be highly sustainable, they are still at the starting stage of understanding the problems and the data related to food waste.

Food waste researches:

Two food waste researchers who have worked on retail food waste in Sweden were interviewed. The main aim of these interviews was to understand the current situation of food waste and how the spoilage sensor could help prevent meat wastage. In relation to the current status of food waste, food waste researcher 2 did not seem to agree with the comments of the store manager and store employee 3. The reason for this was mentioned that if such decrease in food waste from past years had occurred there would have been a lot of discussion in the research

community both to understand and study these solutions for waste reduction. Additionally, frequent dumpster divers discovering edible meat packages from retail store bins was mentioned as a reason. Both researchers mentioned that it is possible that the store employees see it as little waste when compared to total food waste and if actual recorded numbers were taken into consideration they would be large enough to be considered as a problem. This is understandable from literature data as well where the meat department contributed to only 3.67% of the whole wasted mass.

In relation to solving food waste, researcher 2 mentioned focusing on products with low turnover, as they have a higher amount of relative waste. This is because low turnover products due to short shelf life will have less time to be sold. However, as per the literature study Eriksson (2015) mentions that this is also related to the minimum order size of the product. When the minimum order size is larger than required and the product has a low turnover the waste would be high, this was observed when the difference between organic and conventional meat products was studied. Researcher 1 mentioned that food waste solutions need to focus on solving the root cause of the problem to create a substantial waste reduction. The spoilage sensor was considered an interesting approach to waste reduction. However, it was discussed that the waste reduction potential of such a sensor might be limited and that its real waste reduction potential is too difficult to calculate both in the retail and end consumer stages as multiple factors affect the amount of food wasted.

5.2 The Case of a Baltic Retail Chain

The limitations with available retail food waste data, mainly, lower representativeness due to all studied stores being of the same discount store format and the unavailability of data in relation to the cause of waste has led to the need for a case study in a retail chain. Additionally, the different perceptions of quantities of meat waste among the interviewees has supported this need for a case study. While the intention was to consult Swedish retailers and perform the case study due to practical reasons that was not possible. Therefore, a top Baltic retail chain which had agreed to provide data was chosen as the case company to understand the magnitude of waste due to expired due dates. While this data cannot be representative of Sweden it still provides valuable insights related to meat waste. Data for all store formats where fresh meat was sold was collected from the case company and analysed. This data is from store records where employees scan the item being thrown away and register using preset cause of waste. As the organisation is centrally controlled and strict store operations were followed the supply chain employee mentioned that this data is highly reliable. The interviews with the case company employees showed that the general operating procedures in stores were similar to that of Swedish retailers. However, the markdown procedure was centrally controlled by providing department managers a specific budget usable for the markdowns. The procedure regarding the return of lower quality goods to

suppliers was also mentioned to be robust and make sure there are no false claims. The ordering system is also similar to Swedish retail as they are automatic, however manual changes to the order are not common as in some stores in Sweden. Learnings from the interviews of the case company are presented next.

The case company in its multiple retail stores sells different fresh meat products both of the companies' brands referred to as private label products and meat products of other brands. The company is involved in the development of a private label product in more depth compared to that of the other branded products. It all starts with the assortments team where suppliers are first assessed to see if they meet the product and quality requirements of the case company to develop the private label meat products. Suppliers/producers are assessed by evaluating the quality certification and documents along with the ingredients, shelf life tests for the product. The ones who pass these requirements are then tested for taste of the product, upon which a decision is made regarding which private label brand the particular supplier will be used for on an additional basis of the brand criteria such as no additives and no artificial sweeteners. While quality certification is a must for international suppliers, Baltic suppliers without quality certification are accepted if they pass the audit done by the case company. After the decision regarding the supplier is made, a product specification template is sent to the supplier. This template filled by the supplier contains all the food information for consumers mainly the ingredients list, storage specifications and the shelf life. The shelf life is determined by the suppliers, while the case company does not do tests on shelf life themselves, they review the documents related to it. However, in their quality process the case company does make checks for pathogens in the products delivered to them. After the product specifications are finalised the design team develops and sends the packaging design to the case company, once the design is approved by both parties the product is produced and pushed into the supply chain.

Products from all the suppliers of meat are sent to the company’s centrally operated distribution centre (DC). It is at the DC where quality tests for the products are done before they are sent to the stores. The quality checks include visual appearance, packaging and labelling, shelf life and temperature checks. When the products are received at the inbound area of the DC each batch which arrives in multiple pallets are checked before they can be sent into the meat chamber, colour codes are used to mark the status of the pallet. The inbound area for meat deliveries and the meat chamber are maintained between 0℃ to 2℃, while the outbound area where products are loaded onto store delivery trucks is maintained at 4℃. However, products are moved from the meat chamber only 20 minutes before loading. The visual tests are basic checks to ensure that there are no damaged packages, the packaging design and the information about the product are next checked to make sure they are as agreed with the supplier. Similarly, the shelflife of the products has to meet the agreed upon range, with most contracts having ⅔ rd of the total shelf life to be available at the retail store some exceptions based on the suppliers do exist.

Temperature checks of different levels are then performed. First a laser thermometer is used to check the overall temperature of the pallet by taking multiple readings, if this does not satisfy the requirements a digital thermometer is used to take multiple readings around the centre of the pallet. When this test also fails, then the quality controllers are called for who perform an invasive temperature measurement. Products from multiple pallets are selected and meat inside the package is tested, only if this test is passed will the product move forward in the supply chain. Though the maximum pallet weight is 850 kg package sizes itself can differ from product to product and to make sure enough samples of the products are tested before making a decision on rejecting the goods, at least 5% of the delivered products are tested, to have strong evidence even more products are tested in some cases.

Products which do not meet the quality requirements are sent back in case of local suppliers, but in case of international suppliers the products are moved to the incident area before a decision of return or destruction of product is made along with the supplier, in general if the products are of private label then they are destroyed. In all cases the issue is registered with pictures of both the product and the thermometer showing the temperature. Quality issues that are raised by a store after products are delivered are handled in a similar way. When a store makes a quality claim (other than temperature issues), first stores with the same batch of the products are contacted, if more than five stores have similar issues the claim is approved. Before the products are destroyed such claims are reviewed and once the supplier agrees with the decisions a penalty is charged to the supplier. In cases where the temperature of products does not meet requirements at the store and it is evident that the change has occurred from DC to the store the logistics provider is charged for the damage. However, in cases where there is evidence of the products being mishandled at the DC the case company bears the cost. Furthermore, in cases of products not meeting quality requirements a root cause analysis is done, in the former cases the suppliers are asked for an explanation and corrective actions that they will take.

The main causes of waste at these stages are mentioned as products that are received already damaged, for example insufficient adhesive on the package, punctures etc. Products arriving with wrong information printed on them is another major cause, for example the quality team lead mentioned they recently had a supplier send products with the top of the packaging design of one product and the information on the bottom of another product. Temperature differences were mentioned as another cause, trucks stopping during the transport and opening the doors, malfunctioning of equipment in trucks, possible electricity problems in the DC at the meat chamber were mentioned as the causes for the same. Additionally, damage caused due to personnel while loading and unloading the products, while using forklifts to move products within DC are mentioned as causes of waste. Apart from these, outbreaks and quality recalls due to pathogen bacteria can occur. Though these causes were mentioned, they were mentioned as not occurring frequently. As there are strict measures in place to stop such damages from

occurring and having audits every month, inter company (performed by other country branches) audits every year these types of waste are kept to a minimum. Furthermore, the biggest cause of waste in stores was mentioned as the passing of due dates.

The case company has three store formats which sell the majority of different assortments of fresh meat. Hypermarkets, Supermarkets and Convenience stores. All these formats mainly vary in size and the type of assortment. Internally these formats are divided into sub formats of small, compact and larger. While these stores are clustered in such a way the sold assortment varies from store to store as the location of the store defines the customer profile of the store in turn defining the assortment. The daily store operations start with receiving goods from DC. Temperature checks and visual inspection related to damage are done in the store. These packages are then stocked up in the shelves before which products that have passed due dates and do not meet visual quality standards are removed by store employees. While products with medium to long shelf life use an application to suggest critical articles closeby to the due date, fresh meat assortment is manually checked for on a daily basis. The ordering of goods is done automatically, replenishment schedules differ from stores with higher turnover having deliveries six days a week to lower turnover stores having deliveries four days a week. The forecasting and replenishment department is responsible for making the right orders and adjustments to the orders, the store managers are not involved frequently. To ensure the correct orders are made the store has to register all the waste in the system before the end of the working day in order to update to the real inventory status. A preset system with reasons for write-off is used where products are scanned and then a reason is assigned to it before it is thrown in the garbage. These reasons mainly include passed due dates, damage to the product. In stores with kitchens some products can be used as raw material to make meals. Markdowns/ discounts are also used as strategies to push sales of over-stocked and products nearing the due date. Every department in the store gets a markdown budget that is centrally decided by the sector manager and approved by the operations director. The store manager is allowed to use the markdown budget to give discounts in different magnitudes for different products. However, for overstock from campaigns or when a larger number of goods needs to be discounted the decisions are made centrally by allowing for store specific campaigns. The campaign times were also mentioned as the critical times where more wastage occurs both due to overstocking and cannibalization of similar products. Furthermore, since the change in local government regulations on donation the company is working towards partnerships to donate food that is wasted now. However, the sale of goods past their due dates was never heard of.

The overall opinion of the spoilage sensor was positive in all the interviews, the interviewees described the company to be open to innovation and that their teams work towards process and product improvement. The interviewees mentioned that the quality team will have to approve the sensor before it can be used on the company's products. The quality team lead firstly raised

concerns regarding the reliability of the sensor. Secondly, it was mentioned that an extension in product shelf life from producers will only be approved if they provide certifications and test results regarding the same and customer safety will be given priority. Thirdly, as mentioned by all interviewees there would be a need for introducing new standard procedures and training of employees about how to read the sensor and what decisions to make especially when the conventional label and the sensor label show different state of the product. The latter was not considered a hindrance to the deployment of the product as standard procedures for multiple products are followed currently and this would include adding more pointers to the same. The quality team lead and product quality and service manager commented that the cost of the sensor should be low. The latter also mentioned that the sensor needs to be tested on frozen meat that is defrosted and packed as it was a practice by some producers to import frozen meat and then repack it before selling. When asked about the potential value of the sensor the operations director mentioned it could be used for traceability of quality by suppliers. By storing samples of products in an optimal environment from each batch at critical stages in the supply chain they can be used as a measure in case quality claims are made in the store. The operations director also mentioned that the adding cost to the meat product was not a problem as the sensor would be considered as a value adding product and that traceability in the meat segment is something that they are working towards. The product quality and service manager mentioned that it could possibly help in decision making regarding the quality claims which can lead to selling some of the meat frozen rather than as a fresh product. Additionally, the sensor would be a visual cue to quality to the employees. It was also mentioned that this would increase customer trust as the company would be showcasing the quality of their products openly. When asked about the decision making regarding the deployment of the sensor all the interviewees mentioned that there was no standard recipe for such value adding products. The operations director further commented that demanding such additions to products from the retailer will include changes to the purchase price, more about that can only be decided after a pilot test with the sensor. Additionally, it was mentioned that a better case for them would be where the sensor is provided as a product by the producer. Furthermore, the retail association where both retailers and suppliers make decisions was suggested as a place to showcase the sensor and show the benefits to all parties involved in order to make further decisions. Finally, it can be observed that none of the interviewees explicitly recognised the margins used in due dates as a food waste problem and have not seen the sensor explicitly as a food waste solution.

5.2.1 Fresh Meat Waste Quantities at Case Company

The case company data has stores of five different formats viz. Hypermarkets, Supermarkets, Convenience stores, On the go (OTG) convenience stores similar to 7/11 stores in Sweden and low price discount stores. However as of 2019 the Discount store format stores have been converted into the Convenience store format. Therefore, during the analysis the data for the

discount store format has been combined with the data for Convenience stores. An additional reason for the same is that at the time period of the data provided the conversion of many such stores had already taken place. The data provided by the company is from the recent years but before the Covid-19 pandemic struck, as the consumer behaviour during these times would not be representative of the regular times. While the exact years of the data set was not disclosed by the case company the data presented in this report is calculated and presented for a period of 12 months.

The fresh meat products sold at the stores of the case company include minced meat and minced meat products such as burger patties of different meat sources. Products/ different cuts of Beef, Veal, Pork and Mutton both boneless and with bones. Whole Chicken and Chicken products such as legs, wings, breasts. Poultry products such as duck and turkey. The meat departments in all the stores sold 17464.8 tonnes of fresh meat whose monetary value was 56.9 million Euros. Out of this a total of 755.6 tonnes equivalent to 2.356 million Euros was written off/ thrown away in the stores due to various reasons. Out of the total write off 593 tonnes of fresh meat equivalent to 1.893 million Euros was written off as they passed the due date. This is 78.5% of the total written off weight and 80.38% of total written off value. In other words it is 3.4% of net sold weight and 3.33% of net sold value of fresh meat. Table 1 shows the written off values for each product category. While most part of the write-offs are due to wastage from expired due date the remaining small part is due to reasons such as bad quality, damage from DC shipping, damage in store, customer returns, quality recalls, state controlled quality audits and donation to food banks and finally unregistered waste which is the smallest part.

Table 1. Waste of fresh meat per product category

Product Category	**Total Write-off (kg)**	**Due Date Write-off (kg)**	**% Due Date Write-off**	**Total Write-off (€)**	**Due Date Write-off (€)**	**% Due Date Write-off**
Beef minced meat	19499	17607	90.30	78724	70517	89.57
Pork minced meat	14871	13537	91.03	41282	37438	90.69
Mixed minced meat	41854	35546	84.93	132692	111813	84.26
Other minced meat	633	588	92.80	3035	2783	91.70
Minced meat based stuff	14218	12821	90.17	75117	68016	90.55
Chicken minced meat	11854	11262	95.00	34143	31700	92.85

Turkey minced meat	15964	12695	79.52	48124	38871	80.77
Raw sausages	18319	14581	79.60	78618	62465	79.45
Beef products	96747	77098	79.69	386261	352321	91.21
Pork products	260096	194428	74.75	779190	572358	73.46
Mutton products	3132	2898	92.54	29489	26953	91.40
Veal products	7298	6558	89.86	50435	46021	91.25
Whole Chicken	201363	147565	73.28	468179	335458	71.65
Chicken products	21878	20778	94.97	21994	20535	93.36
Poultry	26710	24016	89.91	125880	113804	90.41
Poultry products	1176	1091	92.79	2777	2573	92.66

It can be observed from Table 1 that Pork products, Whole chicken, Beef products and Mixed minced meat are the most written off products in terms of weight. These are also the products most wasted due to expired due dates. However, when looked at as relative waste i.e % of waste due to passed due dates by weight these products appear on the bottom half of the list. Chicken products, Chicken minced meat, Poultry products and other minced meat and Mutton products are the most wasted due to passed due dates. Observations from Table 2 show that Mutton products, Minced meat based stuff, Veal products are the most valuable products. However these are also the products some of the least sold by weight, they follow the same trend in terms of value sold. Minced meat based stuff, Raw sausages, Mutton products and Veal products are the most wasted when looked at as a percentage of sold weight, a similar trend can be seen for sold value. Furthermore, Whole Chicken, Pork products, Mixed minced meat and Beef products are the most sold by weight which in relation to most wasted products can be considered critical.

Table 2. Waste as a part of sales weight and value

Product Category	**Value (€/kg)**	**% of sales wasted due to due dates by weight**	**% of sales wasted due to due dates by value**
Beef minced meat	4.54	4.35	3.84
Pork minced meat	2.92	2.14	2.02

Mixed minced meat	3.74	4.40	3.70
Other minced meat	5.82	3.48	2.83
Minced meat based stuff	6.67	15.25	12.13
Chicken minced meat	3.98	7.19	5.08
Turkey minced meat	3.52	3.95	3.44
Raw sausages	5.39	12.80	10.17
Beef products	5.46	9.92	8.30
Pork products	3.19	3.57	3.30
Mutton products	9.58	12.20	11.85
Veal products	5.86	11.29	13.52
Whole Chicken	2.86	1.93	1.54
Chicken products	1.38	3.95	2.83
Poultry	5.45	5.34	4.65
Poultry products	2.53	4.90	4.57

Table 3 and Table 4 give an understanding of the waste in relation to the different store formats. Product categories that are not sold in all formats except hypermarkets have been removed from the list. Additionally, '-' is used for product categories not sold in that particular format. Observations from Table 3 show that there is no particular trend that is followed by each product category in the different store formats. This means a different amount of waste can occur due to passing of the due dates on products irrespective of the store format. The sales turnover of each product category also varies from format to format. OTG convenience format has the least sales turnover for each product category and is quite low compared to the other formats, for example it has only 0.8 % of whole chicken sales of the convenience format. Similarly, only 4 % of minced meat based stuff sales of the convenience format. The hypermarkets have the highest sales turnover compared to the other formats for all product categories, the turnover is at least 50% higher than that of the supermarket and convenience formats in all product categories. The sales turnover in supermarket and convenience categories however differs from product to product. Observations from Table 4 where the waste is looked into in relation with the sales mostly follow the trend of increasing waste percentage as the sales turnover decreases. However, in certain

cases such as chicken minced meat this trend is not followed and the reason behind this is not clear. Similarly, when looking into the waste in the hypermarkets for different product categories the trend of higher sales turnover products having lower wastage due to passed due dates is not followed, Table 5 presents these results.

Table 3. Percentage(%) of total write off due to passed due dates by weight

Product Category	Hypermarket	Supermarket	Convenience	OTG Convenience
Beef minced meat	91.85	87.62	90.32	87.32
Pork minced meat	94.00	89.29	88.80	91.80
Mixed minced meat	87.96	80.85	83.72	91.69
Minced meat based stuff	94.63	86.91	87.05	97.41
Chicken minced meat	97.28	90.98	93.79	96.03
Turkey minced meat	76.76	79.89	82.91	91.84
Raw sausages	87.22	72.78	77.68	100.00
Beef products	92.14	90.87	59.81	94.12
Pork products	64.21	88.68	86.24	94.42
Mutton products	93.72	81.79	-	-
Veal products	89.73	84.74	99.77	-
Whole Chicken	61.90	87.68	86.88	93.11
Chicken products	95.27	92.29	95.83	-
Poultry	88.95	89.43	94.29	91.30

Table 4. Percentage(%) of total sales wasted due to passed due dates by weight

Product Category	Hypermarket	Supermarket	Convenience	OTG Convenience
Beef minced meat	3.18	4.50	12.77	7.70
Pork minced meat	1.45	2.54	3.84	16.37

Mixed minced meat	3.65	3.73	6.55	10.86
Minced meat based stuff	10.40	14.07	38.75	25.26
Chicken minced meat	5.94	4.73	14.42	28.27
Turkey minced meat	2.76	3.46	13.58	28.46
Raw sausages	8.18	12.17	24.00	87.25
Beef products	7.54	11.95	20.50	31.92
Pork products	2.72	3.15	6.79	17.32
Mutton products	11.51	33.08	-	-
Veal products	10.03	58.60	192.79	-
Whole Chicken	1.61	1.62	3.27	7.60
Chicken products	2.66	3.49	6.05	-
Poultry	4.29	6.75	15.09	5.51

Table 5. Product categories of Hypermarket arranged by due date waste and sales turnover by weight

Arranged highest to lowest waste as % of sales	**Arranged lowest to highest in sales turnover**
Mutton products	Mutton products
Minced meat based stuff	Minced meat based stuff
Veal products	Veal products
Raw sausages	Raw sausages
Beef products	Chicken minced meat
Chicken minced meat	Turkey minced meat
Poultry	Chicken products
Mixed minced meat	Beef minced meat
Beef minced meat	Poultry

Turkey minced meat	Pork minced meat
Pork products	Mixed minced meat
Chicken products	Beef products
Whole Chicken	Pork products
Pork minced meat	Whole Chicken

5.2.2 Potential Saving in Retail and Price of Sensor

As explained in the Industry context chapter, the fresh meat supply chain is characterised by several complex factors. These also make the calculation of the potential savings complex to quantify. As these factors such as consumer demand in terms of type of product and remaining shelf life of product are variable it is not possible to completely estimate the potential. However, there are studies which use mathematical models and run simulations to try and estimate benefits of similar food waste prevention solutions. Broekmeulen and Donselaar (2019) present using two concepts viz. Fresh case cover and efficient frontier which they theorise to be the concepts to quantify the improvement potential that a 43% waste reduction can be achieved when the shelf life of fresh products is increased by one day. However, this calculation is based on a case where all the consumers follow a first in first out rule while purchasing products (FIFO). Grunow and Piramuthu (2013) present theorems and prove that a similar solution i.e using RFID systems to track package level information would impact all stakeholders positively. However, as these are theorems they do not provide any quantification of the potential savings. Buisman, Haijema and Bloemhof-Ruwaard (2019) perform simulations to understand the effect of dynamic shelf life and discounting policies at the retailers. A positive effect for both these solutions is seen on waste reduction, different amounts of reduction in waste and increase in profits is seen in different cases. Wang and Li (2012) also perform simulations and obtain A similar result showing that more profit can be made when a dynamic shelf life is used along with an optimal pricing/ discounting strategy. Tromp et al (2012) who also perform simulations in different scenarios present that a dynamic shelf life is superior to the static labels so much so that in the base scenario the product losses are almost totally diminished and the sales opportunity losses are decreased by 80%. These different studies not only restate the complexity but also show how simulating an ideal case is different from the real scenario. As such mathematical modelling is not in the scope of this research a rather frugal way is opted to understand the effects of these savings on the price of the sensor. It is simple enough to understand that to use such a solution the retailers would need the cost of the solution to be less than that of the potential savings or profit. The following calculations are made to estimate the possible prices of the sensor.

The case company retail stores sell fresh meat in two ways. One, as pre packed goods of different weight denominations, which differs from product to product. Two, as goods that are packed in store as per the customers requirement, these are delivered to the stores in bulk packages. Each package would require one spoilage label, which means the total number of packages sold would be equal to the number of spoilage labels required. Data from the case company provides the number of packages sold for pre packaged meat. In order to make the calculations the bulk packaged meat is converted to a number of packages by using the same relation the total number of pre-packaged units has with their total weight for each product category. This assumes that even the meat packed at the store is sold in the same weight denominations as the pre packed meat. Additionally, products which are only sold by packing in store i.e mutton and veal products are assumed to be sold in denominations of 500 grams and 1000 grams, with 50% of total weight sold in each denomination. Rounding off after adding the number of pre-packed units and assumed number of units for store-packed meat gives a total of 3,77,80,786 units or 37.78 million units. This is the number of labels that will be required over the year.

Table 6. Possible price of sensor

Total number of labels	**100% due date waste reduction value (€)**	**80% due date waste reduction value (€)**	**50% due date waste reduction value (€)**	**30% due date waste reduction value (€)**
3,77,80,786.46	18,93,624.30	15,14,899.44	9,46,812.15	5,68,087.29
Possible price per label (€ cents) →	5.01	4.00	2.51	1.50

Table 6 shows the price per sensor label depending on the quantity of meat waste that can be prevented. The due date value presented in the table corresponds to the cost price for the retailers or the selling price for the producers of the meat packages. This means the retailer would not only save the value corresponding to the waste but would gain the profit after selling the goods that have been saved from waste. Calculating this profit increase is not possible as data in relation to it is not available. Therefore the economic savings have been used to calculate the limits to the price of the sensor. Additionally, it is important to note that retailers do not only gain economic benefits from the sensor but also provide better food safety to their consumers, furthermore, this prevention of meat waste would bring them closer to their annual sustainability targets as meat is the highest contributor to the carbon emissions. Therefore the calculated cost limits of the sensor label need to be taken with a grain of salt. Another observation is that the cost limits calculated would be different from the ones presented in table 6 if each product

category is calculated separately. As the value, amount of waste and the number of sold units differs from product to product some products would allow for the cost of the sensor to be higher than the other products. While this allows for a pricing strategy where the cost of the sensor label is customised based on the product, further analysis would be required to verify its advantage over the single price of label for all product categories. Finally, depending on how the sensor is deployed into the market the burden of the cost of the sensor can fall on different stakeholders, a possibility of sharing these costs among two or more stakeholders is also valid.

5.3 The Business Models

5.3.1 Value Proposition

Before the different business models are presented an understanding of the value proposition offered by the sensor label to the different stakeholders i.e producers, retailers and end consumers are looked into in detail. Due to the different legislations involved with the best before and use by date products certain value propositions differ for each of them which are presented accordingly. It is assumed that the meat products will have both the conventional date labels and the sensor labels on them as a change in legislation where only sensor labels would be enough is not plausible in the near future. Furthermore, these labels provide the option for producers to decrease the safety margins on the dates they use in turn increasing the printed shelf life of the product. However, since the actual margins used today are not known and if producers are not willing to extend the printed life it is possible that some products will have extended shelf life while others will not. Therefore, four scenarios viz. Non-extended Use by products, Extended Use by products, Non-extended Best before products and Extended best before products would arise where certain value propositions will differ as per the scenario. These value propositions are presented in table 7 and next explored in more detail.

Table 7. Scenario Wise Value Proposition

Stakeholder	**Value proposition**	**Use by date products**		**Best before date products**	
		Non-extended	**Extended**	**Non-extended**	**Extended**
Producer	Increased brand value due to increased safety	✔	✔	✔	✔
	Increased brand value due to increased Sustainability	✘	✔	✔	✔
	Increased time for	✘	✔	✘	✔

	processing				
Retailer	Longer time to sell products (increased profit and chance of sale)	❌	✔	✔	✔
	Economic savings from waste prevention	❌	✔	✔	✔
	Increased brand value due to increased safety	✔	✔	✔	✔
	Increased brand value due to increased Sustainability	❌	✔	✔	✔
End Consumer	Increased food safety	✔	✔	✔	✔
	Economic/ Emotional savings due to waste prevention	❌	✔	✔	✔
	Convenient safety indicator	✔	✔	✔	✔

Producers:

Adding the sensor label to their products would enable meat producers to provide an increased amount of food safety for their consumers, this would basically increase their brand value. As the market demand shifts towards sustainability adding the sensor label would help the producer move towards these sustainability goals in turn increasing the brand value. The quantification of these qualitative values is complex, however, we can understand that the potential for savings from waste would be comparatively higher in BBD products. It is also important to note that while these waste savings would occur mostly at the retailer enabling the waste prevention by adding these sensor labels is what would provide the increased brand value to the producer, end consumer perception regarding this should be further analysed to establish what amount of brand value increases for the producer in relation to that of the retailer. In the scenarios where the product shelf life is extended a part of the extension would be available for producers, if current trade practices are followed ⅓ rd of the extended shelf life would be at the producers disposal. The value here can be perceived in two ways, firstly, the producers would be able to decrease the amount of returns from the retailer related to short shelf life deliveries or would be able to avoid discounts due to delivering shorter shelf life products than agreed upon. Additionally, all rejections could be converted to discounts in case of short shelf life deliveries as the actual shelf

life of the product would be known, this is beneficial as selling something is better than selling nothing. Secondly, in cases where the former is not currently perceived as a problem the extra time would still be appreciated as buffer time in cases of machine breakdowns, product reworks, etc. Quantification of this value would require more insight and data from the producers, data from retail or wholesale distribution centres where such decisions on the products are made. As mentioned by one of the interviewees, producers could also use the sensor labels to generate more clarity on product liability during rejections from retailers stating quality issues. This can be achieved by storing a sample of the manufactured batch in optimal storage conditions at the producer and when the case of rejection arises the stored and delivered products can be compared. This would provide confirmation as to whether there was a problem in the manufacturing phase. However, the producer will not be able to identify the exact cause of the problem if it occurred during transport and delivery. While this is a very much plausible value due to lack of insights from the producers regarding the current practices used to identify root causes of quality issues, this research suggests this as an avenue for further research and does not consider it in the business models.

Retailers:

The sensor label seems to provide the most value for the retailers as it would allow them to decrease waste and increase profit. Similar to the producer, BBD products are the ones to which most value is added. In the use-by products where the printed shelf life is not extended, retailers would not gain more days to sell the product i.e there will be no difference from the current waste situation. However, in all other scenarios there would be different amounts of waste reduction. In use by date products with extended printed shelf life retailers would gain more time to sell the product, this will create more chances for a package of meat to be sold. This same trend would be seen in the BBD products. Additionally, the sensors label would enable the retailers in case of non extended BBD products with a choice to sell (maybe by applying discounted price only after the due date which would lead to overall profit increase), upcycle or donate the products after the printed date has passed. This is a change from the current situation where retailers do not see these as options in light of food safety liability. Similarly, in the scenario of extended shelf life products retailers will not only have more chances to sell the product due to longer life but will also have the option to increase profits on discounted products i.e more possibility to sell the product at full price. It is to be understood that waste prevention and extended period of selling go hand in hand i.e as the meat package is prevented from throwing away it gets a longer chance to stay on the shelf. This is the reason why both BBD and Use by extended products would behave the same way in terms of value added by the sensor label. Furthermore, the extension of shelf life would have an indirect positive effect on the on shelf availability which is mainly dependent on the ordering systems and algorithms used by the retailer. In cases where on shelf availability is improved retailers would decrease the lost sales. While quantifying this is complex, results from the case study provide an understanding of the

saving from waste prevention and it is plausible that the impact on shelf availability would be proportional to the current shelf life and the increase in shelf life. Apart from these tangible values other soft values such as food safety and sustainability are gained by the retailers. Providing safer products would increase the brand value of the retailer and waste prevention of meat would bring them even closer to the set sustainability targets as meat has the highest carbon footprint among food products. It can also be noted that meat at this stage would have an even higher carbon footprint i.e both the production footprint and the cold chain logistical footprint. While the reduction in carbon footprint can be quantified similar to the potential reduction of waste the increase of brand value due to this is highly dependent on customer needs and demands. This means the importance of these soft values would differ based on the geographical region's consumer behaviour. Additionally, as the retailer would enable the end consumer to prevent premature meat wastage at home the brand value will increase.

End Consumers:

The main benefit from the sensor label to the meat consumers is food safety. Both during purchasing in store and during consumption consumers will be using this spoilage indicator to make a better decision. Similarly, household meat waste due to passed expiry dates can be decreased as premature throwing away of meat can be prevented by relying on the indicator. Not only will consumers benefit economically by preventing food waste but certain types of consumers will also feel enabled to lead a sustainable life and be free from negative emotions due to wasting. There would not be any waste reduction at households in the scenario of non extended use by date products as these products are recommended not to be consumed by food authorities. In cases of extended use by date products though the products are recommended not to be consumed after the date the sensor label enables printing of the longer shelf life in turn reducing the household waste. In all the other cases the household waste reduction is possible as the meat stored optimally would be indicated good for consumption by the sensor even if the printed date is passed. It is the end consumer's behaviour regarding food safety and that of sustainability that will determine the magnitude of increased brand value for both the retailer and the producers. Similar to other value propositions quantifying these soft values is not in the scope of this research, however, it can be understood from Björklund and Letellier (2020) that food safety is a important value driver for consumers in Sweden but household waste prevention is not seen as a important value driver. This could be due to the fact that consumer's perception of the household waste on an individual level is biassed, data from literature shows that the amount of avoidable meat waste per capita is lower than 28 kg per year in Sweden which means the amount of potential savings per year might not be enough to motivate them to buy only products with the sensor labels on them. Only those consumer segments who are educated regarding the waste problem and its consequences would be attracted to such value propositions. Additionally, this behaviour and the number of people with this behaviour would change regionally making these values more important in some regions than others. Finally, though such sensors are not currently

used it is plausible to state that in comparison to the other digital alternatives where RFID or NFC have to be used to read the shelf life a easily accessible visual alternative would be preferred while shopping in stores.

5.3.2 Value Creation and Value Delivery

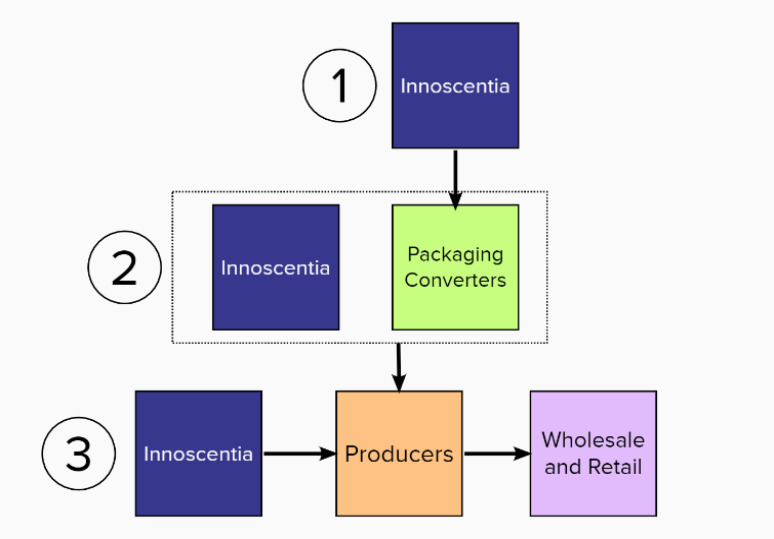


Figure 4. Different ways/cases of value delivery

A business performs key activities to create, deliver and capture value and there are many ways of doing so. Similarly, Innoscentia has multiple options to choose from to build a profitable and scalable business model. The choice of selecting the way value is delivered and to which stakeholder it is delivered to? In other terms the stage of the FSC the company is placed in affects the design of the business model and the results obtained from it. Figure 4 shows the different FSC stages at which the company can establish itself. These are referred to as cases 1,2 and 3 in table 8 where different scenarios in relation to different ways of value creation emerge. The ink developed by Innoscentia is the key ingredient which reacts to the meat atmosphere to change colour and indicate spoilage. This ink can be printed on labels which can then be placed on each meat package. The other option is to use the method where the ink or label is an integral part of the meat packaging film which can be used directly in the packaging line hereby referred to as full film. In case 1 the company would be selling either the ink or the labels directly to the packaging converters who then sell the solution to various meat producers. In case 2 the company sells full film to the producers alongside the packaging converters in other words competes by establishing itself as a player in that industry. In case 3 the company sells directly to the producer but it sells individual label units in other words as a complementary solution to the existing packaging converters. This would also mean selling the technology required to apply the labels by integrating into the packaging lines. In all these cases while the ink is manufactured in

house, the company has a choice to choose between outsourcing printing and printing in house. Additionally, in case 2 an option of collaborating with packaging converters can be explored, in case 3 the label appliers/machinery is developed in collaboration with existing machine suppliers.

Table 8. Different scenarios of value creation and value delivery

Case 1	Selling the Ink	Inhouse manufacturing
	Selling Labels	Outsource Printing
		Inhouse Printing
Case 2	Selling Full film	Outsource Printing
		Inhouse Printing
	Strategic partnership to sell full film	
Case 3	Selling labels and Label appliers	Outsource Printing
		Inhouse Printing

The position of the firm in the supply chain in case 1 is comparatively far from the customer compared to the other cases. While selling directly to packaging converters provides the technology channels to enter the market in a comparatively faster time period there are more limitations to this approach. The distance from the customer due to the "middle man" can imply restrictions in terms of market reach. This technology would be just another part of the whole portfolio of products offered by the packaging converter meaning it will be up to the packaging converter to deploy and diffuse the technology giving them more bargaining power. Additionally, marketing aimed at food producers will still be required from the side of Innoscentia to create a market pull. Based on the understanding from the case study the price of these labels already needs to be quite low indicating profits only at a larger scale. In case 3 this scale required would become even larger as the existence of a middle man logically decreases the margins on the sensor label. In the long term if Innoscentia wants to develop such inks for multiple food products as their core business this still would be a value delivery approach to look into, however if Innoscentia wants to establish themselves on the market within fresh meat alone cases 2 and 3 look more promising. Additionally, as such sensor products are not already in use the burden of creating the market falls on Innoscentia which is key before developing similar inks for other food products. Based on the above reasoning case 1 is not further analysed and left for Innoscentia to pursue, case 2 and 3 are analysed further and the presented business models are based on these cases.

Table 9. Cost structure in different scenarios

	Full film Inhouse	**Full film Outsourcing**	**Labels Inhouse**	**Labels Outsourcing**
Costs Involved	Ink Manufacturing	Ink Manufacturing	Ink Manufacturing	Ink Manufacturing
	Packaging film	Printed film and label		Printed labels
	Label material		Label material	
	Film printer			
	Label printer		Label printer	
	Printing unit and Inventory	Inventory*	Printing unit and Inventory	Inventory*
	Logistics costs (Ink delivery and material delivery)	Logistics costs (Ink delivery)	Logistics costs (Ink delivery and material delivery)	Logistics costs (Ink delivery)
	Personnel		Personnel	

The choice of printing in house or outsourcing the printing in both cases 2 and 3 makes a basic economic difference leading to different profit potentials. Table 9 shows the different costs that would be involved in performing these activities. The in house printing of labels attracts more capital investments than the outsourcing case as there would be a requirement of purchasing equipment, renting units and employing personnel which would also include hiring, training and benefits costs. In the outsourcing case all this work would be outsourced meaning all these costs would be covered while purchasing from suppliers along with the suppliers profit margin, however, the basic difference is at a smaller scale especially in case 2 outsourcing might be cheaper. The question of at what scale it becomes profitable to perform these operations in house needs to be answered by Innoscentia firstly based on the suppliers they chose and economics related to it. Secondly, the company would have to develop competencies related to printing in order to perform operations in house which would be a necessity if printing becomes part of the core business. Outsourcing on the other hand would require lesser competency and provides the opportunity to learn from some types of suppliers and build the competency in the long run. Additionally, the risk of reverse engineering the ink would be higher in outsourcing. While this comparison was mainly intra case, the business models discussed below present the inter case

comparison giving new insights for market entry and long run.

5.3.3 Business Model for Selling Full Film

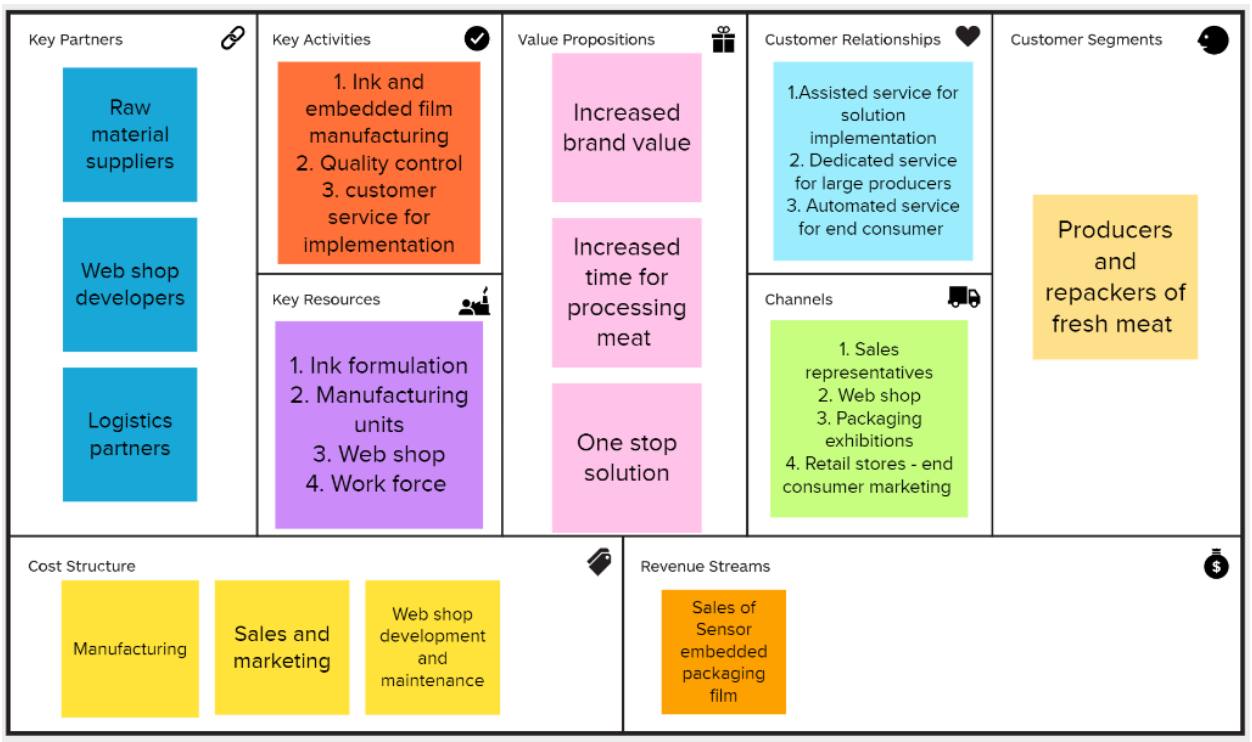


Figure 5. Business model for selling of full film made inhouse

The business model for in house printing in case 2 is presented in Figure 5. The targeted customer segment is the producer of fresh meat to be more detailed it would be the ones producing poultry and red meat. This segmentation is not necessary as there would not be any major difference in the value delivered due to the type of meat, other than the ones mentioned in the value proposition section. The packaging processes of these meats are quite similar with differences in the types of packaging used. The channels to reach these customers would be by use of a dedicated sales team along with the webshop. The webshop and attendance in packaging exhibitions would provide the marketing channel for this customer segment. Additionally, a market pull can be created by getting the acceptance of the end consumer which is achieved by the marketing in retail stores and their channels. It should be noted that the meat packaging itself is a direct marketing channel to the end consumer, further marketing in the retail space can be done using the growing e-commerce channels if necessary. The webshop is a semi automated system which can be used to both attract new clients and manage clients created from the sales team. A portal client specific login in the webshop can allow the established clients to place their order and possibly connect with their ordering systems. This sales team and use of webshop

would build the necessary customer relationships, with the sales force working on a customer acquisition strategy and the webshop mainly on a customer retention strategy. As a new solution in the market the solution's implementation would bring changes to the daily operations of the supply chain players. The transition to these changes should be made as smooth as possible by Innoscentia to show that it is not a hindering factor for the application of the sensor labels thereby leading to the need for assisted customer service. A relationship with the end consumer to create acceptance and provide information regarding the use of the sensors needs to be maintained which can be done through automated service using the company website with the help of bots and FAQs while providing an option for human communication. These relationships also provide valuable feedback from different parts of the supply chain allowing for the betterment of the model overtime.

In order to deliver the mentioned value propositions in the mentioned way a mix of activities would be performed in house and certain activities would be outsourced. Manufacturing the Ink, printing the labels and printing the product information as per the producers request is one of the key activities. Customer service to build the said relationships and quality control mainly due to the number of different orders involved would also be key activities. Development of using this label on different types of packaging should be done in parallel which requires the competency to be developed in house as previously mentioned. Along with the costs presented in the previous section the workforce which includes sales and marketing, R&D personnel, cost of outsourced webshop development and costs of marketing are present. The costs of raw material are dynamic, in order to handle this in the best possible way optimal inventory management and choice of the correct suppliers must be done. Choice of suppliers is also key as this would partly decide the company's serviceability and quality levels. In the same line of argument logistic partners become important to distribute the products on time with no damage caused. A distinction between the type of logistics for shipping the Ink to the manufacturing units and shipping bulk film rolls to producers is made.

The main differences when the choice to outsource manufacturing in this case would be of the cost structure and the key partners as shown in figure 6. Rest of the model would function the same way with differences arising as presented in the previous section. Business model presented in figure 7 however has some more changes which represents value creation through strategic partnerships. Firstly, the outcome of this strategic partnership involves packaging converters to whom the printing of labels, film and their integration is outsourced. This product would then be sold as a collaborative effort having both brand names, for example Innoscentia film powered by XX. The value for the partner would be, access to new innovation and adding it to their portfolio which would also help in increasing their customer base. Multiple partnerships for limited periods can be made based on the geographic markets, the outcomes of the partnership are mentioned here but the details especially in terms of value sharing need to be further developed

by Innoscentia. It should be noted that this model is different from case 1 where only labels are sold to the packaging converter and no clear contact with the rest of the supply chain is maintained.

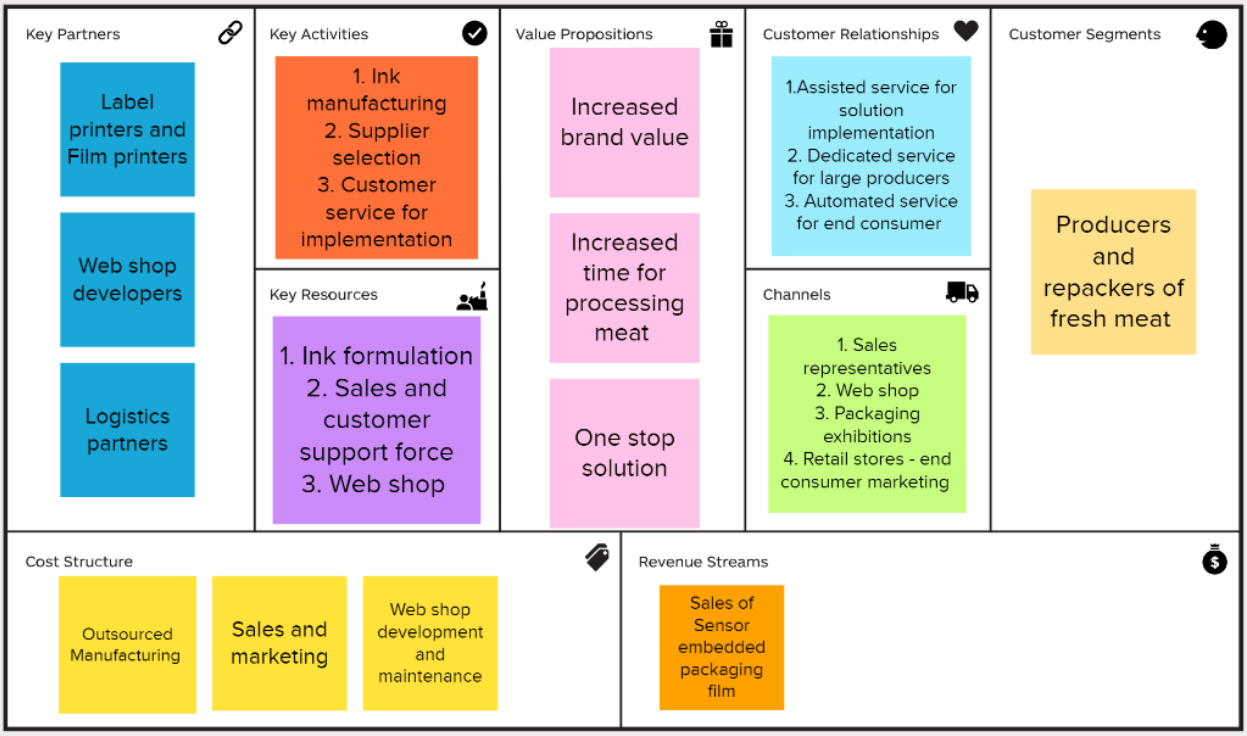


Figure 6. Business model for selling of full film made outsourced

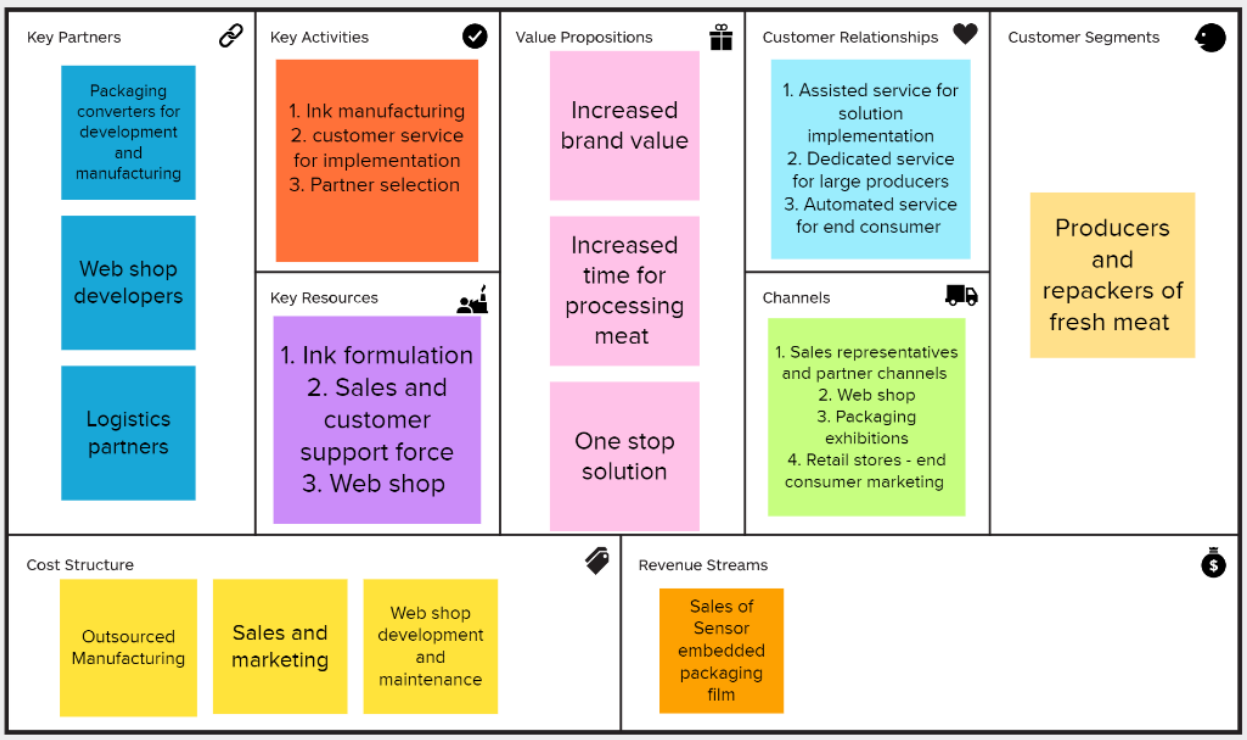


Figure 7. Business model for selling of full film made with strategic partnership

5.3.4 Business Models for Selling Labels

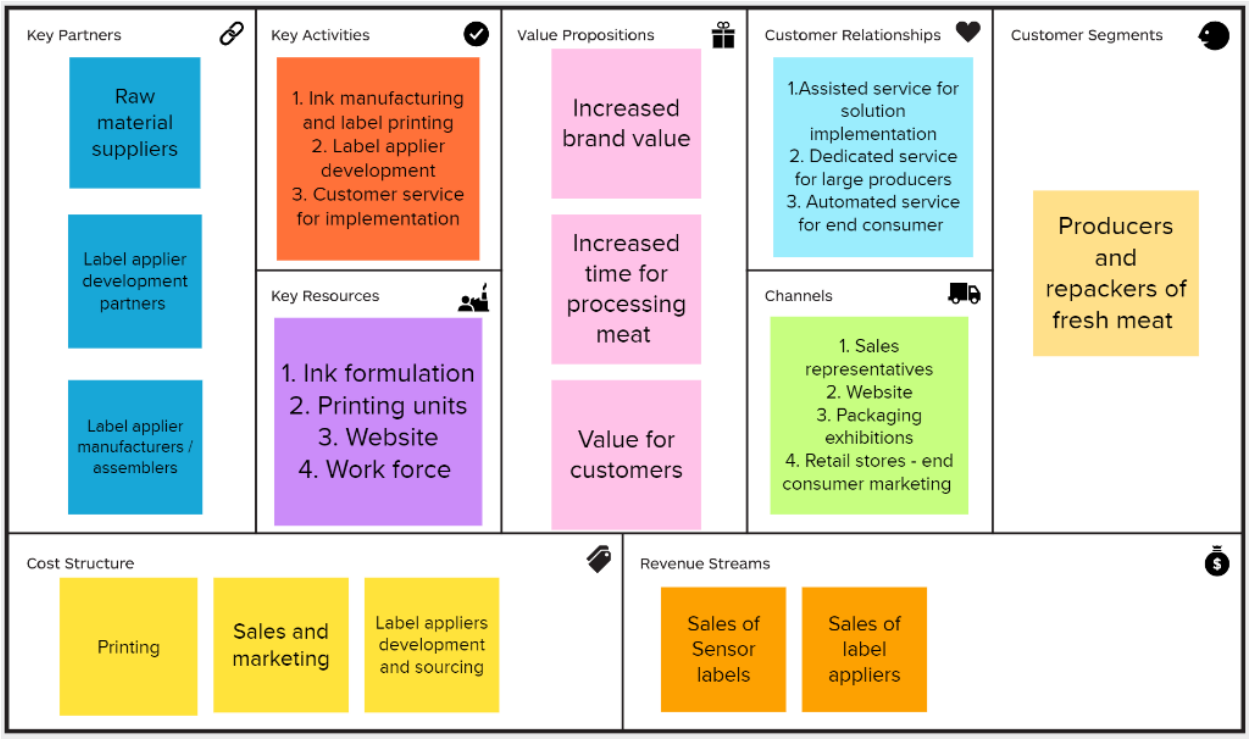


Figure 8. Business model for selling of labels made inhouse

A business model based on case 3 is presented in figure 8. Selling printed sensor labels to the customer segment of fresh meat producers is the main function of this model. Along with the labels a label applier, a machinery that allows to attach the label to different types of packaging is sold to the producers. Though the label printing is done in house the label appliers are developed with the help of consultants; manufacturing and assembly is outsourced. More combinations of this model can be seen if parts of the activities relating to label appliers are shared between inhouse and outsourcing. This would mainly mean a difference in the cost structure and the competence required in the company. The channels to reach the consumers and the relationships to be maintained are not very different from that of the case 2 models. It can be noted that the website is used as a channel instead of a webshop here. While even this model can make use of a webshop a different possibility is depicted here. Website would mainly act as a marketing channel, maintain end consumer relationships and possibly create sales leads. It lacks the functionality of the end to end connection a webshop provides meaning other order processing systems will have to be used. Though this decreases the cost of development and maintenance there would also be change in customer relationship management. Other costs apart from the ones previously presented involve sales and marketing costs, label applier costs. The key activity of customer service would not only be needed for implementation of the new label solution but

also for the integration of the label appliers into the packaging process, post sales service requirements of the label appliers would also need to be fulfilled. The business model if the label printing is also outsourced would remain mostly the same and can be seen in figure 9.

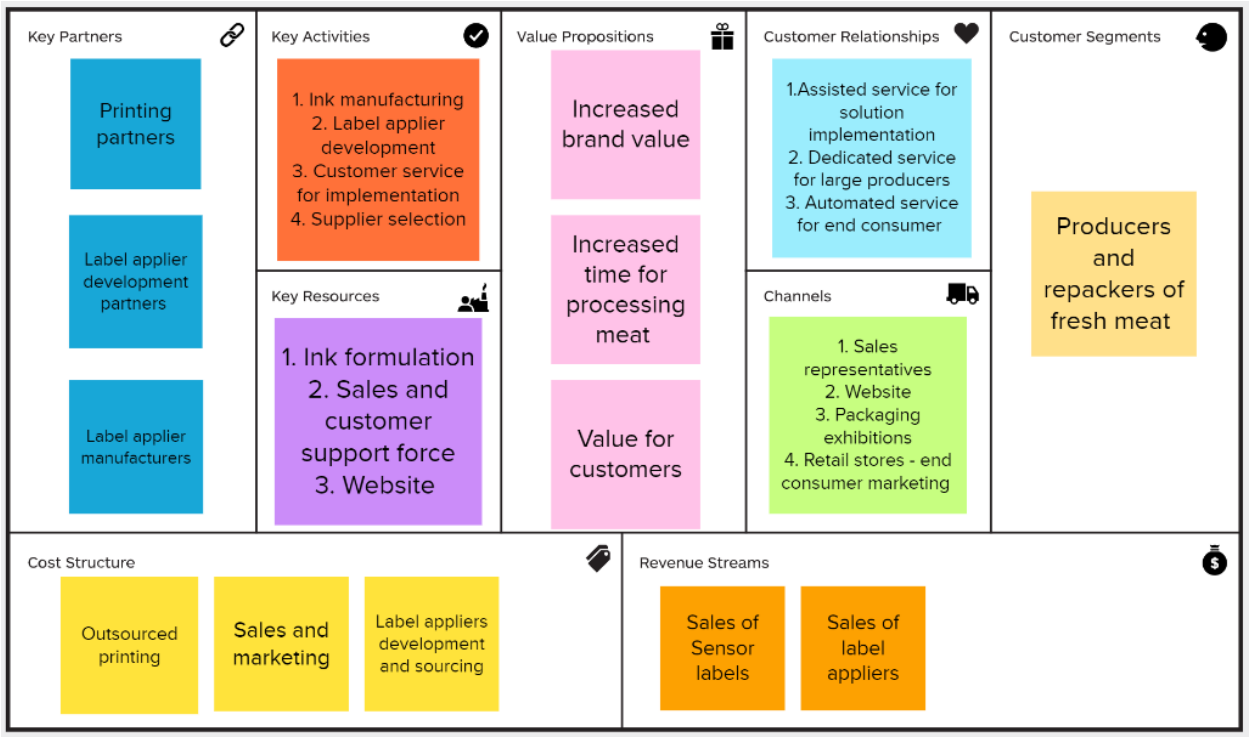


Figure 9. Business model for selling of labels made outsourcing

5.3.5 Comparing the Different Models

All these business models aim to serve the same customer segment but with different approaches to value creation and market position. Firstly, an advantageous value proposition is brought by selling full film represented as the one stop solution in the business models. A full film when compared to the labels would not require the consumer to add extra effort into applying the solution. The investment needed by the customer for the implementation would also go down as label appliers would not be required and the full film can be directly used in existing packaging lines. Additionally, a full film gives the customer a perception of an enhanced product that can substitute his existing suppliers while buying only labels would seem as an extra investment. While these are the indications for the customer the indications for Innoscentia are as follows. Managing the demand and production of full film is comparatively more difficult than that of only labels. Due to different product package sizes, types and different printing content for each producer planning for full film printing is complex and will require strong quality control to reduce returns and rework whereas in the case of printing labels the differences due to size and type of packaging would be minor and will not need printing of different content based on the

brand. This also leads to the understanding that more competencies have to be developed in full film printing in house compared to that of labels outsourcing. The strategic partnership model however, allows to outsource the required competencies from the partners mainly in terms of printing and planning. This model also places itself n a unique place compared to the other models, it requires less capital investment compared to that of in house printing and provides option to learn the required competencies allowing for scaling of the business at a fast rate compared to full film in house which would need lot more investment to scale and might not provide exponential returns at scales smaller to that of bigger players in the market. While the full film inhouse positions itself as a competitor substituting other packaging converters, strategic partnership changes few of those competitors into allies, rightly done this would provide access to the channels of the strategic partners mainly in terms of marketing but also possibly sales. The case 3 models position themselves as a value adding technology but not as competition to the existing packaging converters. It might be of wonder why a strategic partnership is not mentioned in case 3, this is because such a strategic partnership would lead to selling the full film to producers as in case 2 as that would be a better value proposition to offer. In addition to the above comparisons more insights into the product acceptability by the customer, scalability of the models in terms of capacity, market and product, the flexibility and the rigidity of the model which also provides understanding of sustainability, profit potential of the models are presented in table 10. The time and effort required in each model varies and can be seen in these comparisons. The reasoning behind these comparisons is to allow Innoscentia to understand the pros and cons of these choices made, it is to be noted that pros and cons are contextual.

Table 10. Comparison of different models

	Full film inhouse	**Full film outsourcing**	**Full film partnership**	**Label in house**	**Label outsourcing**
Product Acceptability	Higher acceptability due to the product as a single unit.	Higher acceptability due to the product as a single unit.	Higher acceptability due to product as a single unit plus acceptance due to partnership.	Comparatively lower acceptability due to extra investment required.	Comparatively lower acceptability due to extra investment required.
Scalability in terms of capacity	Comparatively difficult due to higher investments needed for scaling	Scaleable due to the minimal capacity constraints, would require more or bigger suppliers	Capacity constraints dependent on the partnership but scaleable by building more partnerships in	Smaller capacity constraints compared to that of full film inhouse. Which allows for	Scaleable due to the minimal capacity constraints would require more or bigger suppliers

			different geographical markets	scaling with lesser investment.	
Scalability in terms of market	Higher competition due to market position would make it comparatively difficult. Establishment in new markets will take more time.	Higher competition due to market position would make it comparatively difficult. Establishment in new markets will take more time.	Competition handled by strategic partnerships. Faster and easier access to the market.	Limited competition as it is a new market with a threat of entry from existing players. Time and effort to access the market would be similar to that of first and third models, however, product acceptability will affect this.	Limited competition as it is a new market with a threat of entry from existing players. Time and effort to access the market would be similar to that of first and third models, however, product acceptability will affect this.
Scalability in terms of product	Same potential to introduce labels into multiple products but more time and effort needed to gain competence of each type of packaging. However, there is a possibility of better quality of products.	Same potential to introduce labels into multiple products. Competencies in terms of different packaging types are outsourced leading to shorter development time.	Same potential to introduce labels into multiple products. Competencies in terms of different packaging types are learnt over time leading to shorter development time and better products.	Same potential to introduce labels into multiple products. Printing Competencies in terms of different packaging types are not needed leading to shorter development time.	Same potential to introduce labels into multiple products. Printing Competencies in terms of different packaging types are not needed leading to shorter development time
Rigidity vs Flexibility	Most rigid due to the choice of in house printing leading to higher investment as a consequence. The same amount of flexibility can be seen in all	Comparatively flexible due to lower investments required. The same amount of flexibility can be seen in all models in terms of value delivery elements.	Partially flexible model as the required capital investments are reduced but rigidity pops in due to the strategic partnerships with	Comparatively flexible due to lower investments required. The same amount of flexibility can be seen in all models in terms	Most flexible due to the least amount of investment required.The same amount of flexibility can be seen in all models in terms

	models in terms of value delivery elements.		lower scope to make bigger changes depending on the details.	of value delivery elements.	of value delivery elements.
Profit Potential	Potential for higher profits at scale compared to model 2 and possibly model 3. Margins on full film can be higher compared to that of only labels as it would include margin on the label and the film. As this is a repeating sales the profit from single time selling of label appliers in models 4 and 5 might not be enough to give more profit in those scenarios.	Similar profit potential of model 1 can be seen here until a scale where inhouse printing provides better margins.	Profit potential is highly dependent on the type of partnership. Supplier bargain power increases due to the partnership and there is a possibility of lower profit compared to that of model 2 due to the same.	Comparatively lower profit potential to that of model 1 even though the two revenue streams exist.	Similar profit potential to that of model 4. However depending on the scale of production one might be more profitable than the other. Comparing inhouse and outsourcing in case1 and case 2 is different as the difference between capital investment is higher. Meaning the scale at which these two models meet the same level of profitability are different.

5.3.6 Other Opportunities and Insights

Retail as a segment:

Apart from the customer segments presented in previous cases, retailers who are a major part of the supply chain can also be considered as a customer segment. There are two subgroups within this segment: retailers who buy meat in bulk and repack it in stores before selling, and retailers who have private label brands and have complete control over packaging decisions. These subgroups, however, represent a relatively small portion of the meat market, which means that the amount of waste caused by expiry dates is also a smaller proportion. Hence, targeting this customer segment alone may not be the key to a sustainable model, it can be used in combination

with other approaches to create a market pull.

The packaging material with embedded sensors can be sold to the first subgroup, retailers who repackage meat in stores before sale. However, the success of this strategy will depend on the acceptance of the end consumer. While food safety is an important consideration for end consumers, it is unlikely that they would be willing to pay extra for a product that they will use shortly after purchase.

For the second subgroup of retailers, who have private label brands and control over packaging decisions, the packaging material must still be delivered to the producers. The sales team may already be targeting multiple producers who are involved in private label packaging, but creating demand from the retailer side by targeting retailers directly could also be beneficial.

As a final option, targeting all types of retailers, including the two subgroups mentioned above as well as retailers who only sell other brands, to mandate the use of sensors on all fresh meat products sold in their stores or to offer incentives for doing so is another potential strategy. A market pull can be created by leveraging the power asymmetry that exists in the food supply chain, particularly in countries where a few retail players dominate the market. This power can be exercised to initiate change, but it is important to carefully consider the way in which this change is initiated. Making such a sale can be challenging, as it may require a significant amount of effort and time, and could potentially impact retailer-supplier relationships

Value addition through upcycling:
The overarching problem in this context is food waste, and the use of sensor labels has the potential to reduce waste to some extent. However, in order to maximise the potential for waste reduction, it is necessary to fully utilise the value of knowing the stage of food spoilage. Both literature and the interview study show that products are not sold past their expiration date in retail chains, even though the law allows for the sale of products that are still safe to consume after the best before date has passed. The use of sensor labels can provide an option for selling products past their expiration date based on food safety considerations; However, consumer behaviour may not support this strategy, which means that unsold meat, even if discounted, may still go to waste, particularly during promotions.

To address this issue, Innoscentia could propose an upcycling program to retailers. This would involve offering a purchasing agreement to buy all unsold meat with one shelf life day remaining, based on sensor readings. The purchase price would be negotiated in such a way that retailers would not incur losses. This would theoretically introduce a zero waste policy in the retail part of the supply chain (although waste would still occur at a systemic level). Innoscentia could create strategic partnerships for the logistics and reverse logistics of the meat from

multiple retail stores of several retailers. A local food processing partner could then upcycle the unsold meat into frozen meat products, which could be re-introduced into the market as a new product, either competing with or substituting other frozen meat products. While this is a very ambitious solution for reducing waste in retail, if successful it could create a strong market pull for the other business models discussed and provide an opportunity for Innoscentia to create a "lock-in" effect with retailers, further securing the business from threats and competition. In addition to reinforcing the other models, this could also be a potential new revenue stream, although its profitability would need to be further analysed. The main costs involved in this model would be logistics, the meat itself, processing, and packaging. The upcycled meat products could be sold with a "waste saved" branding to charge a premium, or they could be sold alongside other products if costs permit. In order to understand retailer acceptance of this model, it may be useful to consider the alternatives to unsold meat. In some larger stores, unsold meat may be used as raw materials in ready-to-eat meals made in-store, but this may not be the most effective solution as it does not add much shelf life to the product, resulting in minimal waste reduction. Other alternatives, such as donation, may also incur losses for retailers. Innoscentia's role in this model would be to bring the different pieces together through strategic partnerships, including the sale and marketing of upcycled meat, in order to focus on its core business of sensor labels.

In order perform further analysis of this model, the following questions should be considered:

- What products should be chosen for this program?
- Who are the local logistic partners who can work with demand uncertainty (as waste generated shifts over time) and perform immediate deliveries?
- Who are the local food processors who can both develop and manufacture the frozen meat products?
- The economics of the model, What will be the cost of the total upcycling process?
- Is there a customer segment that can be targeted to price the products in a way that is profitable?
- Will this act as an additional revenue stream for the company, or only as a market pull strategy? Are the returns from the use of this strategy significantly higher than in its absence?
- Finally, What is the ecological cost of upcycling ? Is it better to upcycle the meat than to waste it?

It is important to consider these questions in order to fully understand the potential benefits and challenges of this model and to determine whether it is a viable option for reducing food waste in the retail sector.

The free rider problem:

In order for the sensor label to enter the market, it must be reliable and gain acceptance from

stakeholders. However, successful commercialization will require overcoming additional challenges. The acceptance of the existence of the value of the sensor label by stakeholders has enabled the company to invest in research and development and reach the current stage of development. However, when it comes to implementing the solution with actual costs involved, supply chain actors are unwilling to share risks and costs, instead waiting for other players to bear the costs while still benefiting from the solution. This behaviour, in which actors try to gain value for free from a solution implemented by another player, is known as free riding.

In the food industry, retailers often expect producers to bear the full costs of implementing solutions like sensor labels, which are applied at the product stage. It is important to note that when labels are sold separately, the cost includes not only the label itself but also the necessary machinery and additional processing time. This issue was also identified with similar technologies like RFID tags; according to a case study by Sarma (2012), retailer mandates for the use of RFID caused tension in supplier relationships and delayed commercialization. In a case study by Olsson (2012), a biobased temperature sensor for food connected to RFID was analysed and the transition from testing to implementation was examined. The attempts to establish a cost sharing model based on value gained were hindered due to the sensitivity of waste data and the reluctance of actors to share it. Additionally, it is believed that supply chain actors themselves are unaware of the exact quantitative value such solutions can provide and whether end consumers will accept and pay for them. This obstacle is further exacerbated by the power imbalance in the supply chain, which leaves little room for cooperation.

Furthermore, Dainelli et al (2008), show that dominant players in the food industry have not taken responsibility for the supply chain or implemented new solutions as they believe existing packaging solutions and systems are sufficient. However, this perspective has started to shift in Sweden due to increasing focus on sustainability and reducing food waste in recent years, leading to the implementation of sustainability targets by these dominant players. Innoscentia will need to consider this issue and devise strategies to encourage dominant players to adopt their technology. While this research did not provide solutions to this problem, it is plausible that the solution can be found through business model innovation; Through changes to value capture systems and engagement of both producers and retailers, Innoscentia can explore potential solutions.

6. Discussion and Conclusions

6.1 Main Findings

This research aimed to identify the characteristics of supply chain actors and determine the value propositions that an analogue food spoilage sensor can offer these actors as part of a business model that defines value creation, delivery, and capture. The value proposition for the three main stakeholders in the supply chain, namely producers, retailers, and end consumers, depends on the type of date labelling currently in use and producers' decision to extend the printed shelf life. Producers can benefit from increased brand value due to increased food safety and sustainability efforts, as well as increase in the amount of time to process meat in their facilities to ensure timely supply Retailers can derive value from increased brand value due to increased food safety and sustainability, economic savings from waste prevention, and the ability to keep products on shelves for longer periods of time. End consumers primarily value the food safety provided by the label, the economic and emotional savings from reducing food waste, and the convenience of simple visual indication provided by the analogue sensor in stores.

The case study on the retail player provided evidence of the problem of meat waste, which is largely caused by the use of conventional expiration dates. As per literature these dates often have longer safety margins, meaning they expire earlier than the actual shelf life of the meat, leading to waste throughout the supply chain, particularly in retail and at the consumer level. The data from the case company showed that 78.5% of total waste generated from various types of fresh meat was due to the expiration of these dates in retail stores. This data also allowed for the calculation of the potential waste reduction, which could be between 18,93,624.30 € and 5,68,087.29 €, if the case company were to use sensor labels. This further provides implications to the price of the sensor ranging from 5 € cents to 1.50 € cents. There has been a lack of information on waste quantification in the retail industry, particularly for certain product categories, due to sensitivity around the issue. This study, therefore, agreed with some previous research but disagreed with others, due to the limited data and level of detail in the data, as well as the specific geographic region used in those studies.

Based on the understanding gained from the literature study, interviews, and case study several business models for the focal company were presented. The large number of choices that can be made in relation to the business model elements has resulted in a wide range of possible models. This study presented a few combinations of these choices as business models for the focal company. Five business models have been presented based on how the product is delivered (just a label or a full film) and how value is created (by performing activities in-house or outsourcing them) to different customer segments. These business models were then compared to each other

based on their acceptability, scalability, flexibility, and profit potential. Each model has its own strengths and weaknesses based on these attributes. This research does not provide a complete business model that the focal company can immediately adopt, but rather shows the different possibilities that the company can consider when developing its business model. The best choice for the company will depend on its goals, internal factors such as funding, and the changing business environment.

6.2 Practical Implications and Recommendations

Conventional date labels have provided users with the convenience of an end date, at which point the product would diminish in quality. While for end consumers this is an attribute leading to the instore purchasing choice, supply chain actors use it for planning and trade agreements. Indicators that only show the state of a product at a given point in time would not meet these requirements and could create challenges for the implementation of the product. To address this, Innoscentia should consider finding a way to display the estimated date at which the product will no longer be consumable on their analogue sensors. This would allow them to meet the unspoken requirements of supply chain actors and improve the implementation of the product.

Digitalisation and Industry 4.0 are trends that more and more players are investing in to streamline their operational processes. Innoscentia's digital sensor is a good step towards the future, but adding digital elements to the analogue sensor to make it semi-digital could enhance its value proposition. This digital element could be an external device, such as a scanner, that converts the analogue indication to a digital language. This would strengthen the value proposition for supply chain actors, who could use real-time freshness data for demand planning and in-store discounting applications. End consumers could also benefit from this as there is a growing use of IoT applications

BMI is a longitudinal process that requires ongoing efforts to review and improve the company's business model. While this research may provide some initial guidance, it is important for the company to regularly reassess and update its business model and cultivate a mindset of considering the business model in all decision-making. By doing so, the company can remain competitive and adapt to changing market conditions. The focal company has the option to choose a single business model presented, blend elements from different models, or even use two models concurrently. However, it is important for the company to consider potential conflicts and path dependency when making these decisions.

Being a supply chain innovation that provides value to multiple stakeholders, a core function of a successful model would be to maintain healthy relationships with all the stakeholders. It is important to consider the asymmetry in the value gained by different stakeholders, as this can

potentially jeopardise these relationships. To ensure the overall success of the model, it is necessary to consider the needs and concerns of all stakeholders and strive to create a balance of value for all parties involved. It is likely that the retailer who is also the dominant player in markets like Sweden, derives the most value from the sensor label even though the label enters the supply chain at the producer. In order to successfully commercialise this product, it may be necessary to use both push and pull strategies, which may involve the use of multiple models simultaneously.

The value that can be derived from the sensor label is influenced by both the cultural context of the end consumer and the business environment. This means that the company may need to use custom models for each geographic market that it targets in order to effectively reach and serve these customers. By taking a systems view of these models, the company may be able to identify synergies and optimise the performance of the business models.

Along with the competence of formulating the sensing ink, the company will need to build strong competence in forming strategic partnerships, particularly in certain business models.To achieve a global presence it will be necessary to develop a faster process for building these optimally flexible partnerships. In addition to culture, legislation is also a factor that can impact the value proposition of the sensor label. However, changing legislative matters on a global scale is a very lengthy process. To minimise the risk of delays in commercialization due to legal obligations, it is important for the company to begin discussions with legislative bodies along with the competitors about the use of such labels as part of its long-term planning efforts. Considering the timing and funding constraints that may be involved, it is important for the company to be proactive in addressing legal considerations and ensure that they do not hinder its ability to bring the product to market. One key factor that the focal company should consider when working further on the presented business models is the revenue streams. From a long-term sustainability perspective, it is important to consider the potential risks of relying on a single stream of revenue, particularly in the face of change. To mitigate these risks, the company should consider exploring additional revenue streams that could be generated using its existing competencies. By diversifying its sources of income, the company may be better able to weather market changes and ensure its long-term viability.

Food waste is a systemic and complex issue, with various types of food having different characteristics that contribute to the problem. While research has provided some insights into the magnitude of food waste, it is difficult to fully understand the potential of solutions to address this issue because it depends on many factors. The sensor label is just one of many solutions that need to be implemented in combination in order to effectively tackle the problem of food waste. As the trend towards ecological sustainability grows, it is important to consider the negative impact of meat consumption on the planet. When promoting the sustainability advantages the

sensor can provide for the meat product category, the focal company must be careful not to give the wrong impression. It is important to be transparent about the potential for change and not allow other supply chain players to use the product as a tool for greenwashing. The company should focus on upholding its core values and overall goal of reducing food waste to gain the trust of the end consumer.

While some of the recommendations provided in this research may seem obvious or already known to the focal company, it is important to emphasise them because they represent fundamental considerations that are necessary for achieving business success and fulfilling core values. The business models presented in this research are not necessarily radical, but they offer a systems perspective on the key activities and competencies that the focal company should focus on in order to achieve its goals.

6.3 Theoretical Implications and Future work

The food waste quantification done in this research contributes to a specific section of food waste research attempting to quantify the problem. The very limited availability of food waste data on the retail stage makes this research a valuable addition. The specificity of the waste data on fresh meat products along with the quantification of reasons behind waste have not been explored much in extant literature. Each food product category has certain attributes which indicate the risk of waste in the supply chain but a larger part of food waste research has focused on all products together giving a general idea of waste. However, In order to employ food waste solutions and gain success specific categories have to be studied on a more detailed scale. While recent studies have focused on fresh fruit and vegetable and fresh bread categories an avenue of future research is open to study more of these products and limitedly researched categories such as fresh meat.

Gaining a lot of popularity during the internet boom, the concepts of business model and BMI have opened a new dimension of innovation. While the antecedents were of mainly different ways of value delivery due to the possibilities created by the internet and digitalization many other dimensions of BMI have been explored. The existing frameworks and the BMC used in this research provides the nine basic elements for all types of businesses. While the visual representation of the canvas gives a systems view the dynamics of relation between elements is not presented visually. Additionally, as the canvas is made for generalisation focus on elements needed for specific business environments is decreased. Such is the case for businesses pursuing value capture in complex supply chains where the consequences of the business model to the relationship with a supply chain actor is of relevance to study. This provides two avenues for future research. One is the study of characteristics and focal elements of a business model, its effect on firm performance for businesses catering solutions to complex supply chains. Second is

the conceptualization and validation of BMI tools for such specific cases, this could be done by adding elements to the BMC and designing a way to visually communicate relationship dynamics inspired by other frameworks with a lower level of abstraction.

6.4 Limitations

Due to practical constraints and the limited scope of the research, this thesis has not delved deeply into certain topics and has focused on certain areas of interest. This indicates that there is still much more work to be done on the business models and establishing the value propositions. The focus of this research has been on the retail part of the food supply chain, which was chosen due to its significant influence in the supply chain. While the retail stage was studied empirically, the other stages were only examined using existing literature. It will be necessary to further study the end consumer acceptance of the solution and the value created for the producer empirically. Additionally, the data on food waste in this research is based on a single retail chain in one region of the world, which limits the generalizability of the findings. Furthermore, this research has not considered the variability in food waste over time, such as whether certain periods are associated with higher levels of waste. While the overall waste numbers may be reliable, it is not known when the most waste is generated. The calculation of the waste reduction potential and the associated value quantification was done in a simplified manner due to the complexity of accounting for multiple factors, which was beyond the scope of this research. This suggests that while the focal company should put in a reasonable amount of effort into theoretically calculating the value quantification, more practical or empirical approaches may be more effective for studying the value in detail. This could involve conducting experiments or gathering real-world data to more accurately assess the value of the solution and its impact on different stakeholders. The limitations of this research also impact the business models presented, as the research setting and current stage of product development have only allowed for the presentation of qualitative reasoning when comparing the business models. More complex economic calculations would be needed to further validate these qualitative results and provide more detailed insights into the potential viability of the different models.

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Appendices

Appendix I - List of Interviewees

S.No	Interviewee	Length of Interview	Interview Channel
1.	Vice President of a restaurant wholesaler operating in six European markets	35 mins	Phone call
2.	Food regulatory affairs specialist at Livsmedelsföretagen	25 mins	Phone call
3.	Experienced Swedish retail store employee 1	40 mins	Phone call
4.	Experienced Swedish retail store employee 2	20 mins	In person
5.	Experienced Swedish retail store employee 3	20 mins	In person
6.	Store manager of a Swedish retail store	50 mins	In person
7.	Food waste researcher 1	45 mins	Zoom
8.	Food waste researcher 2	30 mins	Phone call
9.	Innovation engagement lead at a Swedish retailer	30 mins	In person
10.	Supply chain manager at case company	30 mins	Microsoft Teams
11.	Private label quality team lead at case company	50 mins	Microsoft Teams
12.	Product quality and service manager at case company	50 mins	Microsoft Teams
13.	Store (Hypermarkets) operations director at case company	50 mins	Microsoft Teams

In addition to the above-mentioned interviews, meetings with the CEO of Innoscentia in person and using zoom were conducted frequently to answer questions regarding the company and their understanding of the market and product. Furthermore, short meetings with the founders of the company and the R&D head have been conducted. Lastly, meetings with an employee from the supply chain department at the case company were conducted in order to help with understanding and collection of food waste data.

Appendix II - Data Request to the Case Company

1. How much of each type of fresh meat is sold? (In number of units and weight)
   a. Amount of meat sold in relation to the number of days left to expire or best before.(Example: How much chicken is sold when it has 6 days left?)
   b. The sales data divided by the type of store
2. How much of each type of fresh meat is thrown away at the stores? (In number of units and weight)
   a. What are the reasons behind the waste generated?
   b. What percentage of the waste generated is due to what reason? (Example: How much is thrown away because the product reached an expiry date?)
   c. The waste data divided by type of store
3. What types of packaging is used for each of the fresh meat products?
   a. How much percentage of each type is used for each product?
   b. What percentage of the cost is the packaging for each product?
4. How much of each type of fresh meat is procured at the warehouse?
5. How much of each type of fresh meat is thrown away at the warehouse? (In number of units and weight)
   a. What are the reasons behind the waste generated?
   b. What percentage of the waste generated is due to what reason? (example how much is thrown away because product reached expiry date?)
6. How many cases of recall have happened in the last 5 years?
   a. How much meat was thrown away due to that? (In economic value and in weight of meat)
   b. What were the reasons for recall?
7. How many times in the last 5 years have the products been sent back to the supplier?
   a. How much meat was sent back/ thrown away due to that? (In economic value and in weight of meat)
   b. What were the reasons for rejection of stock? and What percentage was sent back due to what reason?
8. What is the market share of the company for each fresh meat product?
9. How often do you stock out on different types of fresh meat products?
   a. What is the amount of stock outs divided by type of store?
   b. What is the estimate of lost sales caused by stock-outs?

Questions 1,2 and 5 were completely answered by the case company. Data from question 5 is not used in analysis as the waste quantities due to due dates were minimal at the warehouse compared to the stores.

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